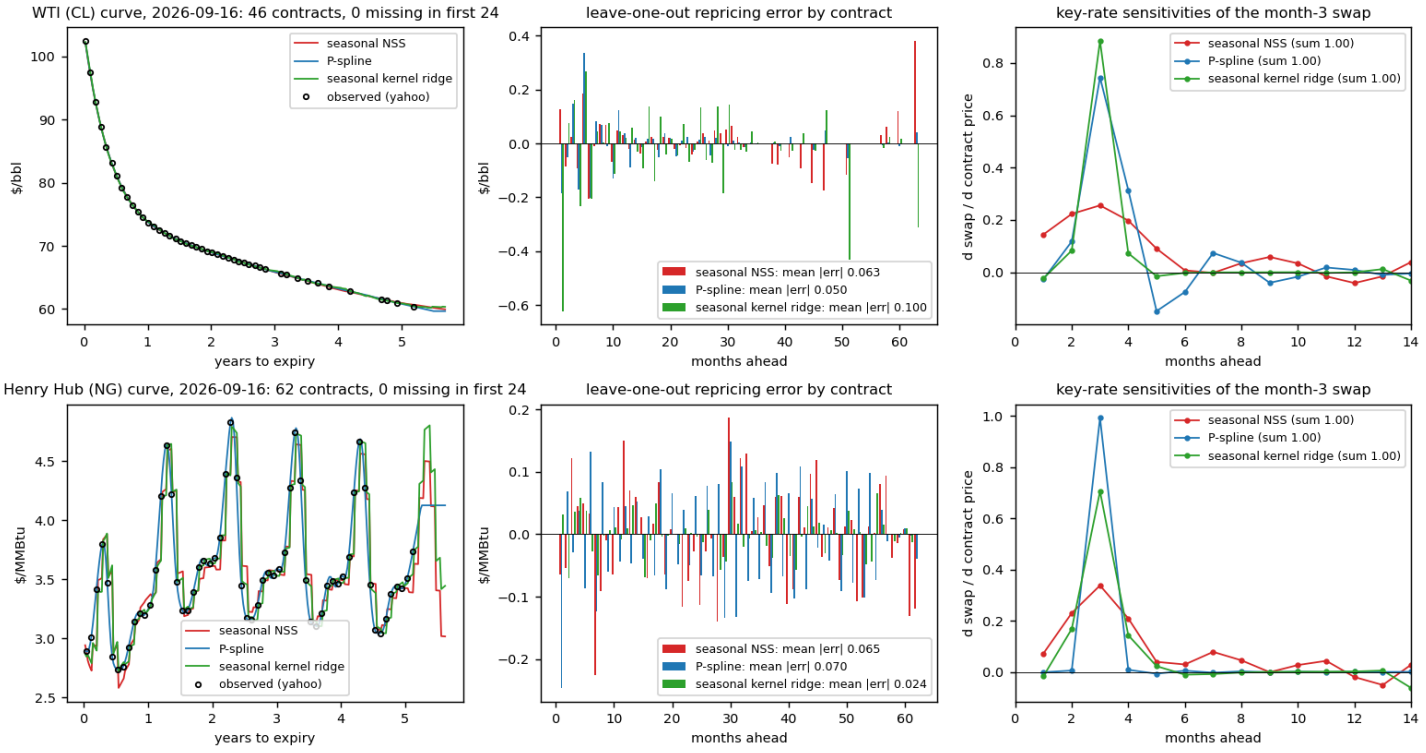
Curves, Impact-Aware Hedging and RFQ Quoting for Commodity Swaps

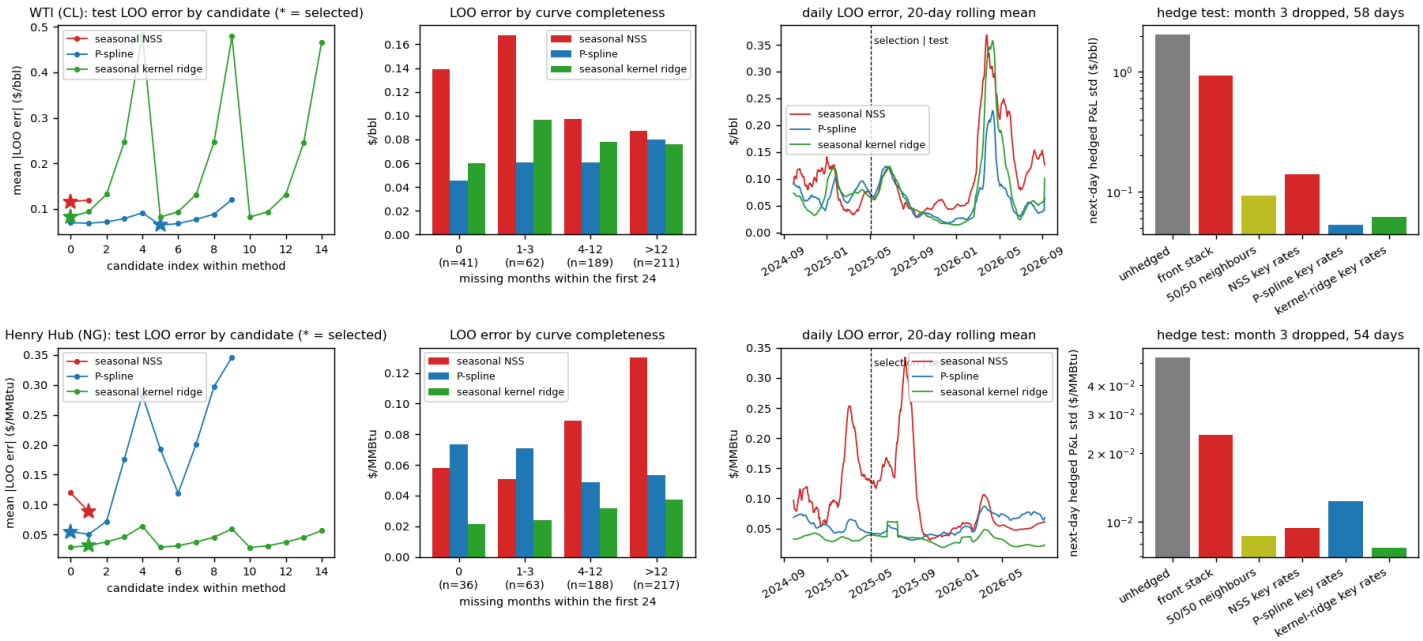
Question. A commodity swap desk quoting calendar-month WTI and Henry Hub swaps by RFQ needs a futures curve to price off, hedge ratios for a book of swaps, a hedge execution policy that respects price impact, and a quoting rule that survives informed flow around the weekly EIA releases. Does the recent research - seasonal kernel-ridge and P-spline curves (Filipovic-Pelger-Ye kernels), nonparametric cross-impact (Hey-Neuman-Tuschmann), impact-aware hedging and the Barzykin-Bergault-Gueant closed-form multi-asset quotes - improve on the desk's classical answers (seasonal Nelson-Siegel-Svensson, front-month stacks and strips, symmetric spreads) when each is tested out of sample on free public data and, for quoting, in an RFQ simulator with an exact P&L decomposition?

Method. Header-only C++ behind one CLI, a FIX 4.4 RFQ gateway with an identical-fills self-test, and a numpy reference implementation. Curve: seasonal NSS, cubic P-spline with a second-difference penalty and a seasonal kernel ridge (Sobolev cubic kernel times a periodic kernel), all linear in prices so leave-one-out repricing, extrapolation and hat-matrix key-rate sensitivities are exact; candidates selected on the first third of the daily builds and scored on the rest; calendar-month average swaps with the NYMEX expiry rules. Risk: PCA of the ladder. Impact: pooled power-law self-impact and ridge cross-impact from an hourly order-flow-imbalance proxy, block-bootstrapped over days, with a cumulative decay curve; hedging policies (front stack, strip, factor-minimum, impact-aware with ablations) compared at equal residual risk on random swap books. Quoting: tiered clients with logistic hit probabilities and a hidden informed direction that drifts the market; symmetric quotes, the closed-form skewed quotes, adverse-selection and calendar widening, an oracle; exact decomposition into spread, inventory, adverse, hedge execution and hedge impact; mark-outs, toxicity routing, anticipatory hedging, risk limits; a replay of the real 2024-2026 days with release-day jumps.

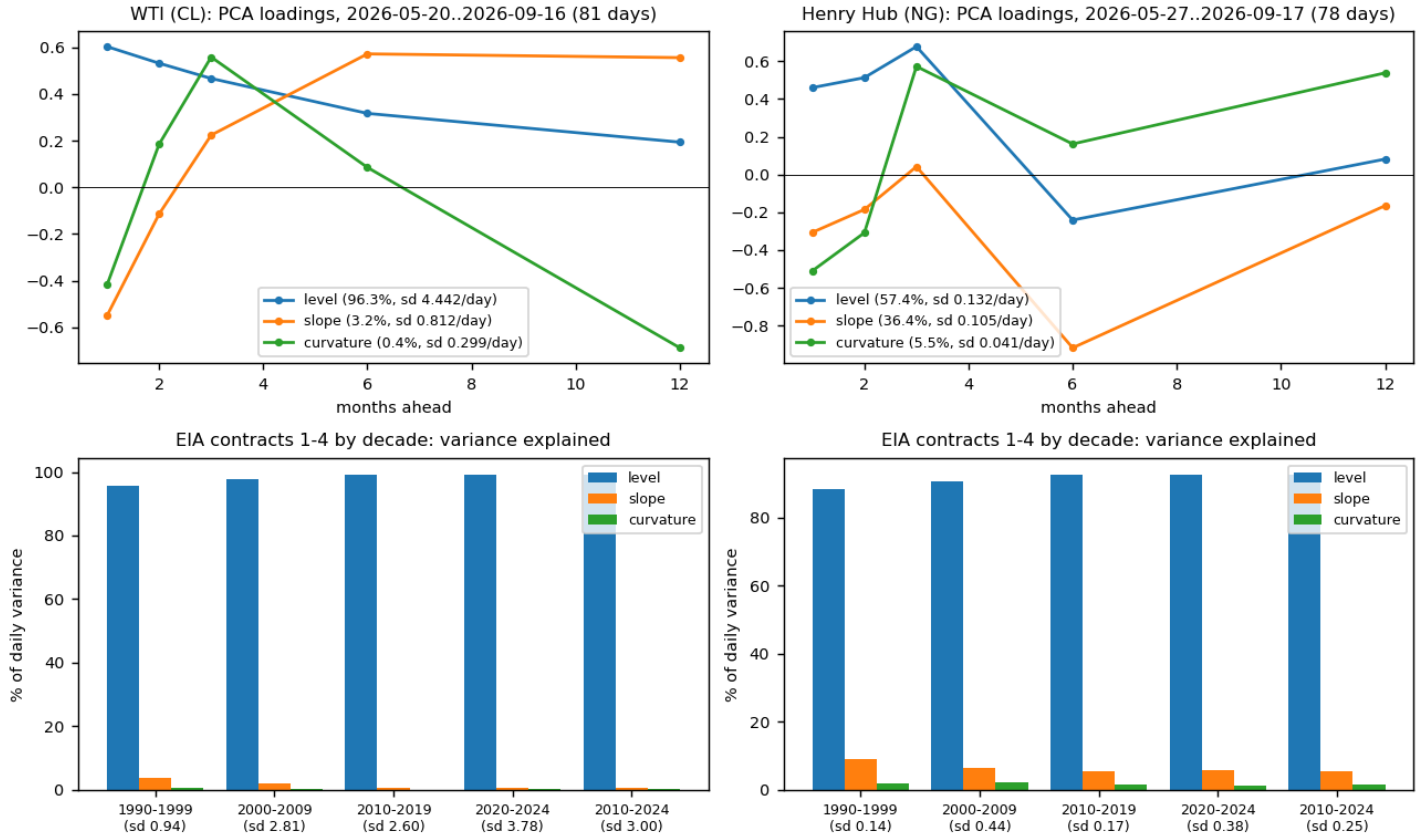
Caveats. There are no free historical full curves: the daily history is the EIA front contracts 1-4 to April 2024 plus two years of listed contracts from Yahoo, so complete curves exist only from July 2026 and every result carries the curve completeness of its day. Client flow is simulated and the impact proxy is order-flow imbalance from hourly bars, not tick data.



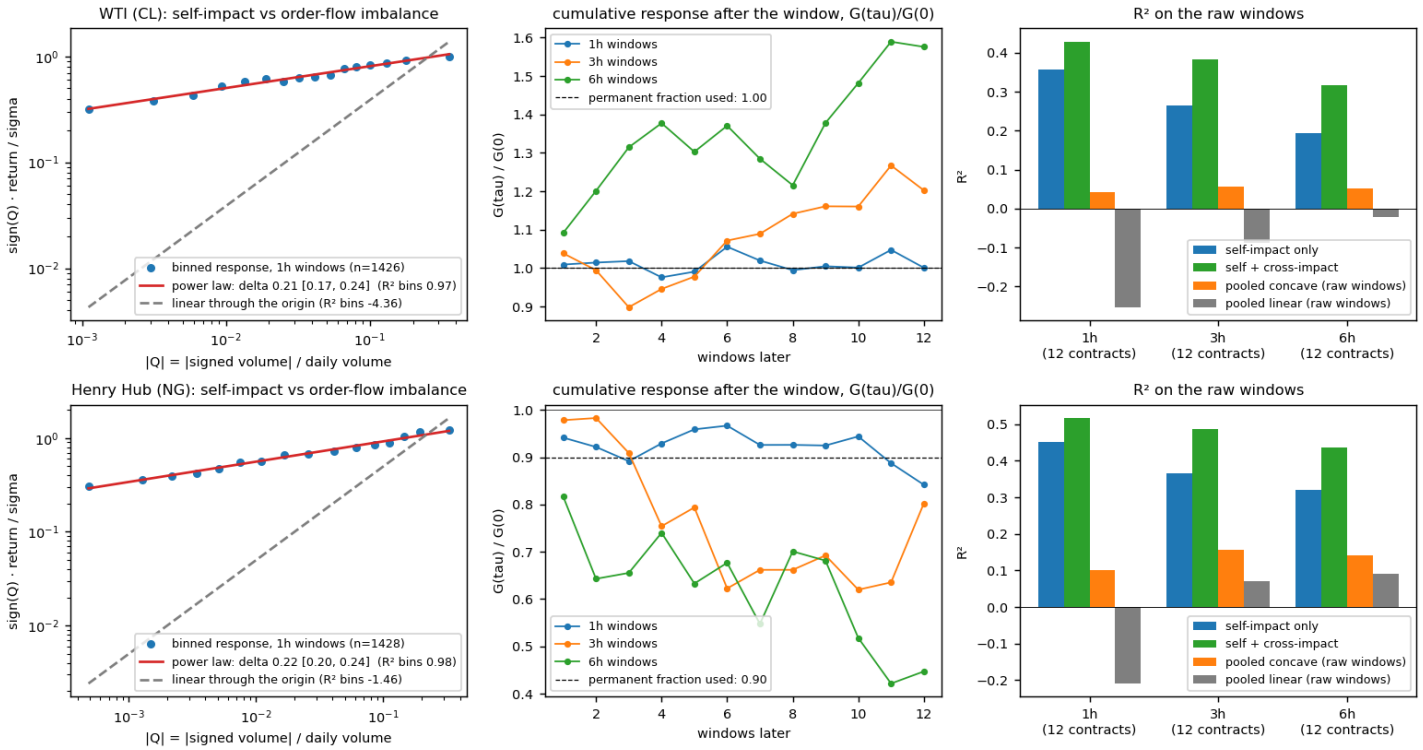
Curve construction on one day: the three models on the observed contracts, leave-one-out repricing errors by contract, and the key-rate sensitivities of the month-3 calendar swap.



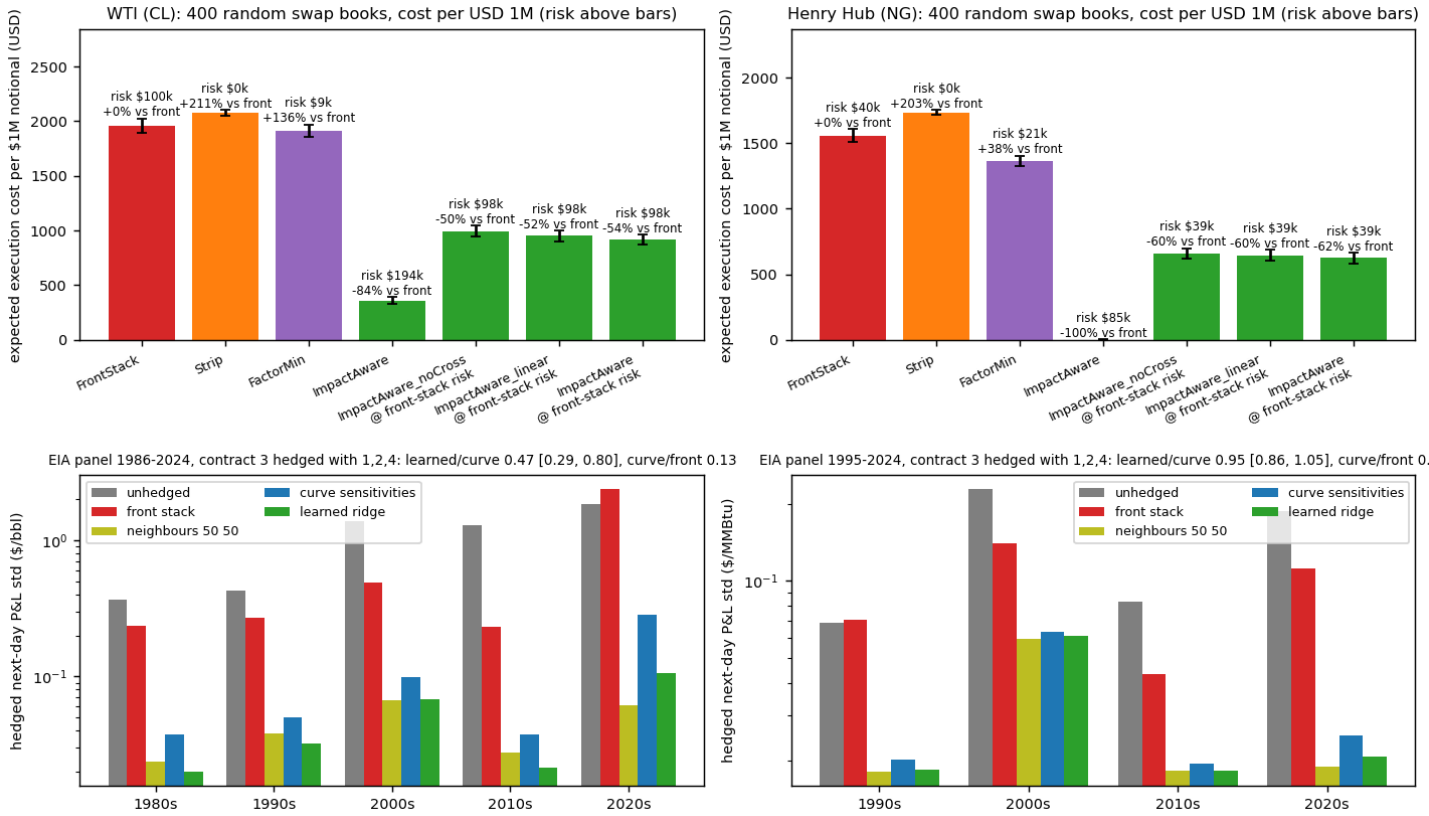
Daily builds: test-period LOO error by candidate with the selection, LOO by curve completeness, rolling daily LOO, and the hedge test (month-3 contract dropped and hedged with each curve's key rates).



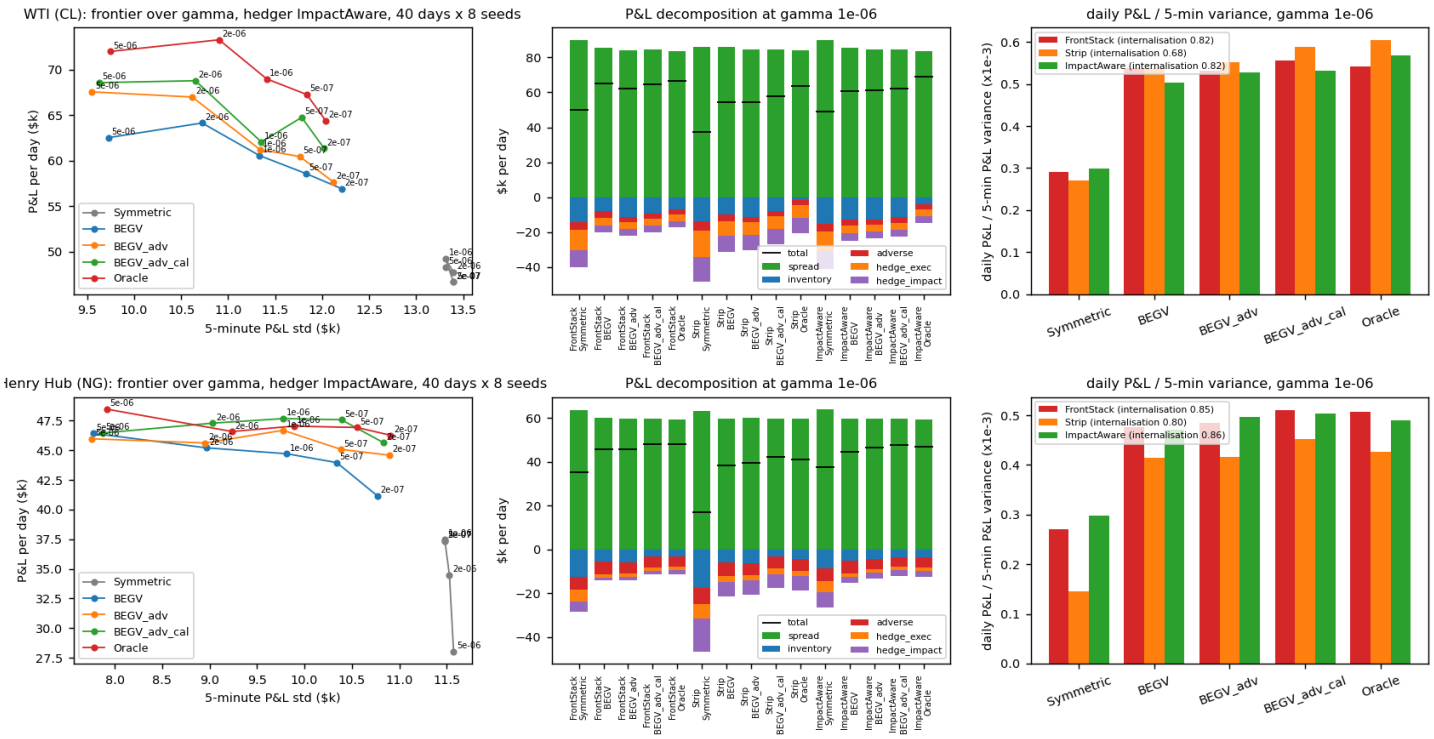
PCA loadings on the ladder over the recent window, and variance explained by decade on the EIA front contracts.



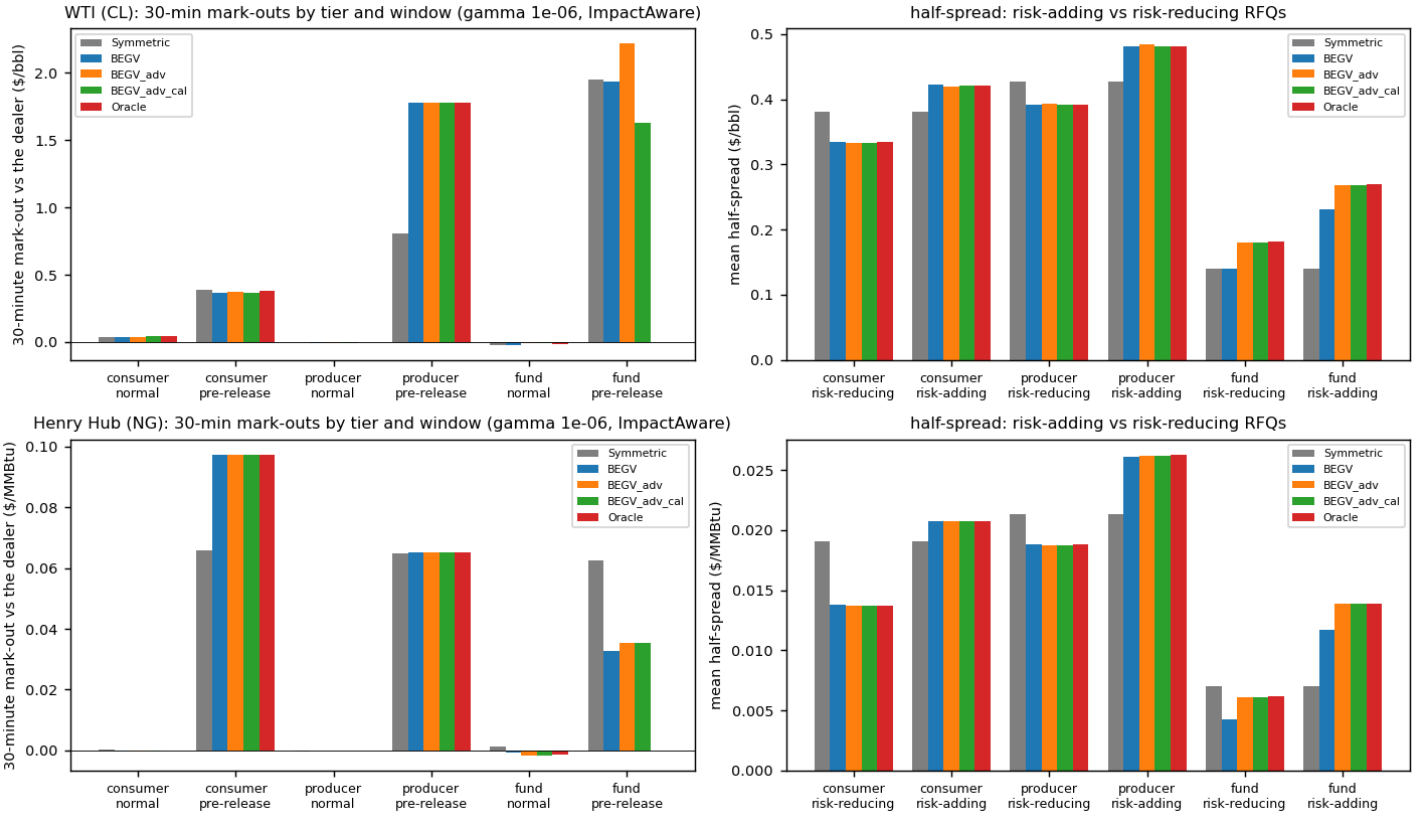
Price impact from hourly bars: binned self-impact with the power-law and linear fits, cumulative decay after the window, and the explanatory power of self vs self-plus-cross impact.



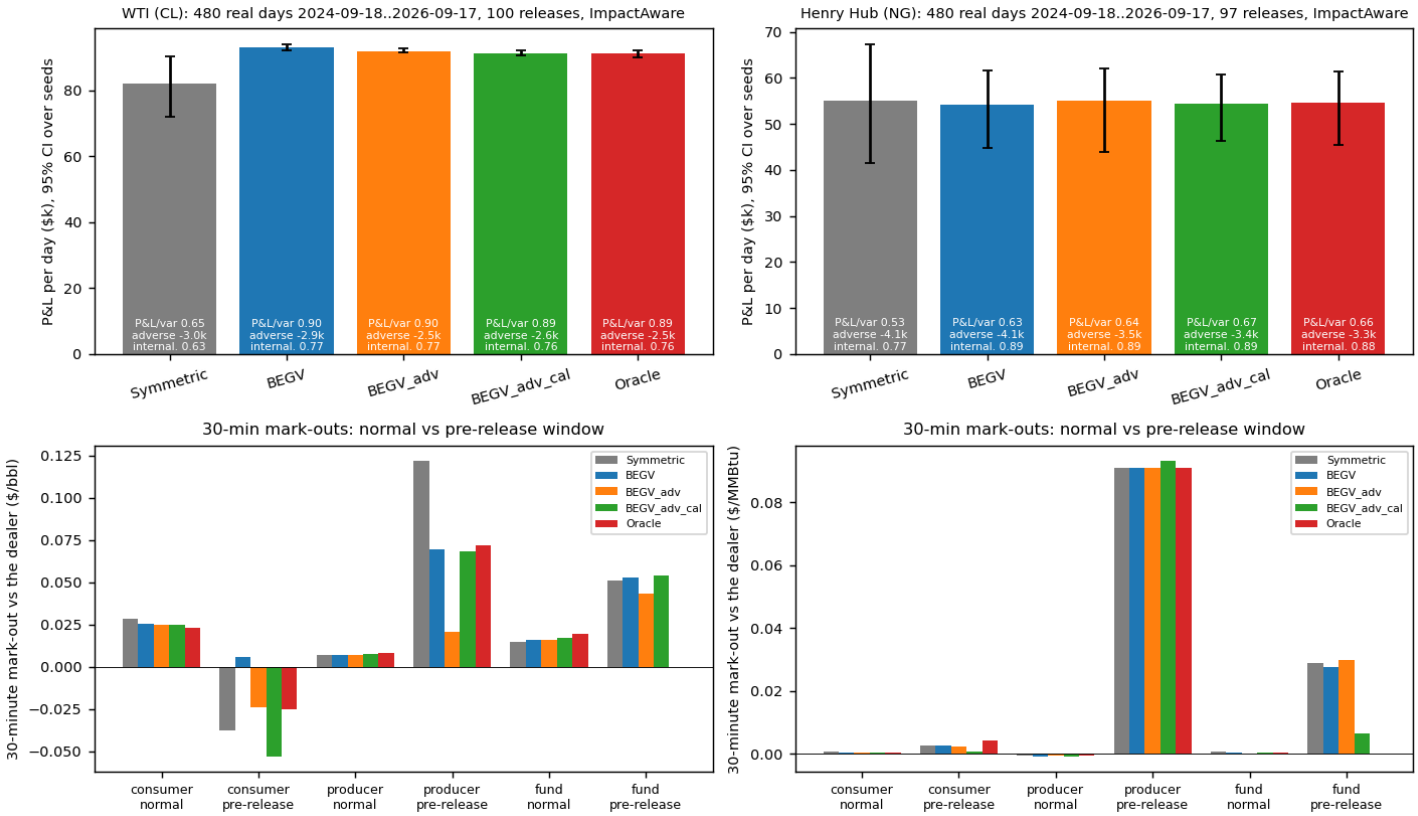
Hedging random swap books: execution cost per $1M at each policy's residual risk and at the front stack's risk (equal-risk ablations); the EIA-panel hedge test by decade with learned vs curve ratios.



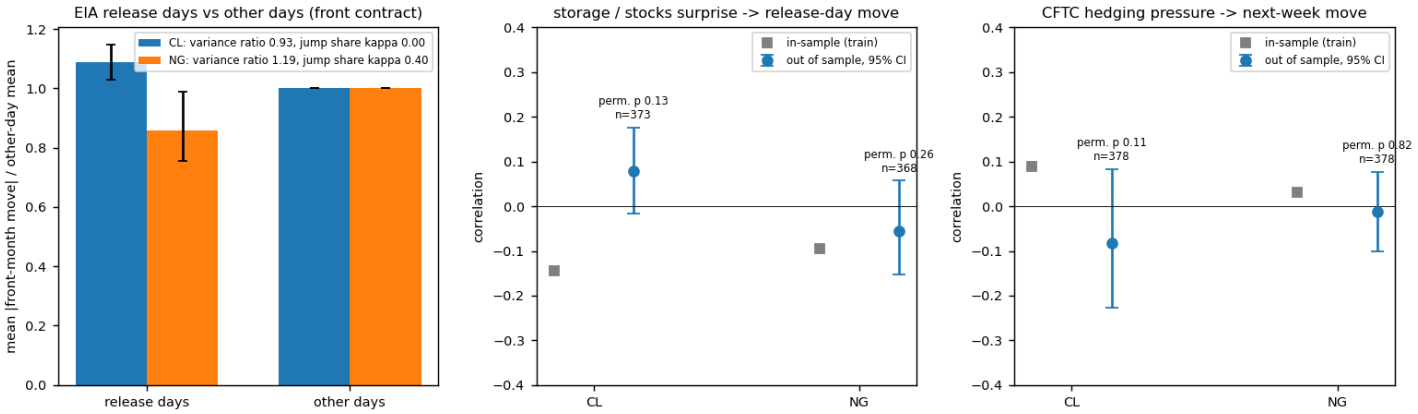
RFQ simulator: frontier over risk aversion, P&L decomposition and risk-adjusted P&L by quoter and hedger.



Quote structure: mark-outs by tier and window, and the skew between risk-adding and risk-reducing RFQs.



Replay of the real 2024-2026 days with EIA release jumps: P&L by quoter with CIs and mark-outs around releases.



Release-day study on the EIA history: release vs other days, inventory-surprise nowcast and hedging-pressure correlations out of sample with permutation p-values.

Generated by scripts/summarize.py from results/*.json. Costs and P&L in US dollars; prices in $/bbl (WTI) and $/MMBtu (Henry Hub).

Data

Curve days by source: EIA front contracts 1-4 (1983/1993 to April 2024), then the listed contracts from Yahoo (daily, two years) merged with the continuous front; the histogram counts the missing delivery months within the first 24 for the days since September 2024.

									root
CL	9661	1985-01-02	2026-09-17	9157	41	62	189	212	2026-07-21
NG	7653	1994-01-19	2026-09-17	7149	36	63	188	217	2026-07-29

Source parity on the overlap days (EIA front contract vs the Yahoo continuous front): CL: 20 days, median |diff| 0.000, 95% within 1% of the price; NG: 15 days, median |diff| 0.000, 100% within 1% of the price.

WTI crude (CL): curve construction on 2026-09-16 (46 contracts, 0 missing months in the first 24)

model	lambda	in-sample rmse	LOO mean abs err	interior	LOO %	extrapolation err (las	3m strip	12m strip	sum of month-3 key rat
seasonal NSS	-	0.0744	0.0631	0.0544	0.090%	+0.162 / +0.231 / +0.4	88.0133	79.5685	1.000
P-spline	1e-04	0.0346	0.0501	0.0472	0.064%	-0.003 / -0.043 / +0.3	88.0133	79.5685	1.000
seasonal kernel ridge	1e-05	0.0175	0.1003	0.0836	0.135%	-0.065 / -0.404 / -1.2	88.0133	79.5685	1.000

Calendar-month swaps off the selected model (business-day-weighted average of the contracts prompt on each day; legs marked * are priced off the model because the contract is not observed):

			month
2026-10	95.5791	22	2026-11 x 0.591, 2026-12 x 0.409
2026-11	91.4967	21	2026-12 x 0.667, 2027-01 x 0.333
2026-12	87.7970	23	2027-01 x 0.652, 2027-02 x 0.348
2027-01	84.7462	21	2027-02 x 0.619, 2027-03 x 0.381
2027-02	82.6500	20	2027-03 x 0.750, 2027-04 x 0.250
2027-03	80.4409	23	2027-04 x 0.652, 2027-05 x 0.348
2027-04	78.6664	22	2027-05 x 0.591, 2027-06 x 0.409
2027-05	77.2886	21	2027-06 x 0.619, 2027-07 x 0.381
2027-06	76.1400	22	2027-07 x 0.682, 2027-08 x 0.318
2027-07	75.0136	22	2027-08 x 0.591, 2027-09 x 0.409
2027-08	74.1973	22	2027-09 x 0.636, 2027-10 x 0.364
2027-09	73.4945	22	2027-10 x 0.636, 2027-11 x 0.364

Henry Hub gas (NG): curve construction on 2026-09-16 (62 contracts, 0 missing months in the first 24)

model	lambda	in-sample rmse	LOO mean abs err	interior	LOO %	extrapolation err (las	3m strip	12m strip	sum of month-3 key rat
seasonal NSS	-	0.0677	0.0651	0.0642	1.872%	-0.011 / -0.144 / -0.1	3.5369	3.2075	1.000
P-spline	1e-04	0.0035	0.0701	0.0676	2.021%	-0.014 / -0.127 / -0.3	3.5369	3.2075	1.000
seasonal kernel ridge	1e-04	0.0156	0.0245	0.0248	0.691%	-0.017 / -0.055 / -0.0	3.5369	3.2075	1.000

Calendar-month swaps off the selected model (business-day-weighted average of the contracts prompt on each day; legs marked * are priced off the model because the contract is not observed):

			month
2026-10	3.0677	22	2026-11 x 0.864, 2026-12 x 0.136
2026-11	3.4691	21	2026-12 x 0.857, 2027-01 x 0.143
2026-12	3.7575	23	2027-01 x 0.870, 2027-02 x 0.130
2027-01	3.3840	21	2027-02 x 0.857, 2027-03 x 0.143
2027-02	2.8275	20	2027-03 x 0.850, 2027-04 x 0.150
2027-03	2.7379	23	2027-04 x 0.870, 2027-05 x 0.130
2027-04	2.7857	22	2027-05 x 0.864, 2027-06 x 0.136
2027-05	2.9543	21	2027-06 x 0.857, 2027-07 x 0.143
2027-06	3.1521	22	2027-07 x 0.864, 2027-08 x 0.136
2027-07	3.2138	22	2027-08 x 0.864, 2027-09 x 0.136
2027-08	3.2117	22	2027-09 x 0.864, 2027-10 x 0.136
2027-09	3.3262	22	2027-10 x 0.864, 2027-11 x 0.136

WTI crude (CL): daily builds 2024-09-17..2026-09-17 (503 days; candidates selected on the first 167 days, sco

model	selected	test LOO mean abs err	test LOO %	vs NSS (paired boots	test extrapolation err	LOO on complete days	LOO on days >12 mi	LOO 2024	LOO 2025	LOO 2026
seasonal NSS	harmonics=2	0.1159	0.160%	+0.0% [+0.0, +0.0]	0.2127	0.1389	0.0873	0.1031	0.0640	0.1636
P-spline	lambda=0.0001 knots=26	0.0648	0.088%	-26.4% [-33.4, -18.4]	0.1066	0.0451	0.0798	0.0683	0.0633	0.0730

seasonal kernel ridge	lambda=1e-05 ell=0.2	0.0825	0.109%	-19.4% [-26.9, -10.9]	0.1517	0.0597	0.0761	0.0536	0.0639	0.1081

Hedge test on the 58 complete days: the month-3 contract is dropped, priced off the curve built from the rest and hedged with that curve's key-rate sensitivities; next-day hedged P&L std in $/bbl (unhedged 2.035):

hedge			
front stack	58	0.9252	54.5%
kernel ridge	58	0.0620	97.0%
neighbours 50 50	39	0.0923	95.5%
pspline	58	0.0526	97.4%
seasonal nss	58	0.1392	93.2%

All candidates (test-period LOO):

method					
seasonal_nss	harmonics=2	0.0831	0.1159	0.0762	0.1572
seasonal_nss	harmonics=3	0.0861	0.1195	0.0789	0.1590
pspline	lambda=0.0001 knots=1m	0.0770	0.0700	0.0631	0.1179
pspline	lambda=0.001 knots=1m	0.0764	0.0688	0.0610	0.1179
pspline	lambda=0.01 knots=1m	0.0820	0.0717	0.0650	0.1180
pspline	lambda=0.1 knots=1m	0.0841	0.0789	0.0666	0.1189
pspline	lambda=1 knots=1m	0.0821	0.0916	0.0665	0.1225
pspline	lambda=0.0001 knots=2m	0.0729	0.0648	0.0559	0.0897
pspline	lambda=0.001 knots=2m	0.0863	0.0679	0.0666	0.0877
pspline	lambda=0.01 knots=2m	0.0868	0.0772	0.0680	0.0810
pspline	lambda=0.1 knots=2m	0.0826	0.0887	0.0665	0.0782
pspline	lambda=1 knots=2m	0.0842	0.1208	0.0746	0.0870
kernel_ridge	lambda=1e-05 ell=0.2	0.0691	0.0825	0.0599	0.1168
kernel_ridge	lambda=0.0001 ell=0.2	0.0724	0.0941	0.0640	0.1152
kernel_ridge	lambda=0.001 ell=0.2	0.0781	0.1322	0.0714	0.1145
kernel_ridge	lambda=0.01 ell=0.2	0.0911	0.2478	0.0897	0.1377
kernel_ridge	lambda=0.1 ell=0.2	0.1342	0.4823	0.1496	0.2945
kernel_ridge	lambda=1e-05 ell=0.35	0.0692	0.0824	0.0600	0.1169
kernel_ridge	lambda=0.0001 ell=0.35	0.0724	0.0939	0.0640	0.1152
kernel_ridge	lambda=0.001 ell=0.35	0.0781	0.1319	0.0714	0.1147
kernel_ridge	lambda=0.01 ell=0.35	0.0910	0.2473	0.0896	0.1380
kernel_ridge	lambda=0.1 ell=0.35	0.1334	0.4788	0.1484	0.2877
kernel_ridge	lambda=1e-05 ell=0.5	0.0692	0.0824	0.0600	0.1169
kernel_ridge	lambda=0.0001 ell=0.5	0.0724	0.0939	0.0640	0.1152
kernel_ridge	lambda=0.001 ell=0.5	0.0781	0.1318	0.0714	0.1147
kernel_ridge	lambda=0.01 ell=0.5	0.0907	0.2457	0.0891	0.1378
kernel_ridge	lambda=0.1 ell=0.5	0.1307	0.4662	0.1446	0.2734

Henry Hub gas (NG): daily builds 2024-09-17..2026-09-17 (504 days; candidates selected on the first 168 days

model	selected	test LOO mean abs error	test LOO %	vs NSS (paired bootstr	test extrapolation er	LOO on complete days	LOO on days >12 mis	LOO 2024	LOO 2025	LOO 2026
seasonal NSS	harmonics=3	0.0885	2.627%	+0.0% [+0.0, +0.0]	0.0555	0.0582	0.1299	0.0897	0.1303	0.0603
P-spline	lambda=0.0001 knots=1m	0.0548	1.560%	-6.3% [-12.3, -0.2]	0.1720	0.0736	0.0533	0.0641	0.0432	0.0684
seasonal kernel ridge	lambda=0.0001 ell=0.2	0.0314	0.874%	-51.7% [-54.5, -48.7]	0.0343	0.0215	0.0373	0.0364	0.0337	0.0291

Hedge test on the 54 complete days: the month-3 contract is dropped, priced off the curve built from the rest and hedged with that curve's key-rate sensitivities; next-day hedged P&L std in $/MMBtu (unhedged 0.053):

hedge			
front stack	54	0.0240	54.7%
kernel ridge	54	0.0077	85.5%
neighbours 50 50	33	0.0086	83.7%
pspline	54	0.0122	76.9%
seasonal nss	54	0.0094	82.4%

All candidates (test-period LOO):

method					
seasonal_nss	harmonics=2	0.1407	0.1196	0.0929	0.0675
seasonal_nss	harmonics=3	0.1218	0.0885	0.0585	0.0513
pspline	lambda=0.0001 knots=1m	0.0558	0.0548	0.0502	0.1766

pspline	lambda=0.001 knots=1m	0.0615	0.0504	0.0521	0.1422
pspline	lambda=0.01 knots=1m	0.0981	0.0716	0.0746	0.1055
pspline	lambda=0.1 knots=1m	0.2319	0.1763	0.1837	0.1368
pspline	lambda=1 knots=1m	0.2926	0.2831	0.2836	0.2191
pspline	lambda=0.0001 knots=2m	0.2100	0.1932	0.1922	0.2089
pspline	lambda=0.001 knots=2m	0.1051	0.1191	0.1066	0.1765
pspline	lambda=0.01 knots=2m	0.2648	0.2010	0.2083	0.2298
pspline	lambda=0.1 knots=2m	0.3117	0.2971	0.2979	0.1593
pspline	lambda=1 knots=2m	0.3385	0.3463	0.3431	0.4139
kernel_ridge	lambda=1e-05 ell=0.2	0.0351	0.0288	0.0299	0.0280
kernel_ridge	lambda=0.0001 ell=0.2	0.0345	0.0314	0.0316	0.0293
kernel_ridge	lambda=0.001 ell=0.2	0.0363	0.0377	0.0353	0.0404
kernel_ridge	lambda=0.01 ell=0.2	0.0394	0.0458	0.0402	0.0519
kernel_ridge	lambda=0.1 ell=0.2	0.0641	0.0640	0.0634	0.0459
kernel_ridge	lambda=1e-05 ell=0.35	0.0350	0.0286	0.0300	0.0279
kernel_ridge	lambda=0.0001 ell=0.35	0.0346	0.0312	0.0316	0.0291
kernel_ridge	lambda=0.001 ell=0.35	0.0363	0.0375	0.0353	0.0404
kernel_ridge	lambda=0.01 ell=0.35	0.0391	0.0455	0.0401	0.0523
kernel_ridge	lambda=0.1 ell=0.35	0.0561	0.0592	0.0561	0.0447
kernel_ridge	lambda=1e-05 ell=0.5	0.0350	0.0285	0.0300	0.0278
kernel_ridge	lambda=0.0001 ell=0.5	0.0346	0.0310	0.0316	0.0290
kernel_ridge	lambda=0.001 ell=0.5	0.0363	0.0373	0.0353	0.0403
kernel_ridge	lambda=0.01 ell=0.5	0.0389	0.0451	0.0400	0.0525
kernel_ridge	lambda=0.1 ell=0.5	0.0525	0.0564	0.0528	0.0441

WTI crude (CL): learned vs curve hedge ratios on the EIA front panel 1986-01-10..2024-04-05 (8907 days, trail

Contract 3 hedged with contracts 1, 2 and 4; next-day hedged P&L std in $/bbl. Learned = ridge regression of contract-3 changes on the neighbours over the trailing window; curve = key-rate sensitivities of the kernel_ridge fitted each day. Learned / curve std ratio 0.473 [0.288, 0.799]; curve / front-stack 0.131 [0.123, 0.209].

							hedge
unhedged	1.1900	0.0%	0.3691	0.4291	1.3728	1.2897	1.8412
front stack	0.8772	26.3%	0.2356	0.2715	0.4881	0.2327	2.3617
neighbours 50 50	0.0470	96.0%	0.0236	0.0380	0.0668	0.0276	0.0612
curve sensitivities	0.1153	90.3%	0.0378	0.0498	0.0981	0.0377	0.2841
learned ridge	0.0545	95.4%	0.0201	0.0323	0.0685	0.0216	0.1059

Henry Hub gas (NG): learned vs curve hedge ratios on the EIA front panel 1995-02-17..2024-04-05 (6898 days

Contract 3 hedged with contracts 1, 2 and 4; next-day hedged P&L std in $/MMBtu. Learned = ridge regression of contract-3 changes on the neighbours over the trailing window; curve = key-rate sensitivities of the kernel_ridge fitted each day. Learned / curve std ratio 0.950 [0.862, 1.048]; curve / front-stack 0.406 [0.330, 0.488].

							hedge
unhedged	0.1596	0.0%	0.0687	0.2274	0.0831	0.1863	
front stack	0.0987	38.2%	0.0703	0.1396	0.0435	0.1118	
neighbours 50 50	0.0371	76.7%	0.0181	0.0595	0.0182	0.0189	
curve sensitivities	0.0400	74.9%	0.0200	0.0633	0.0194	0.0251	
learned ridge	0.0380	76.2%	0.0183	0.0609	0.0182	0.0207	

Factor model (PCA of daily price changes)

root	window	days	ladder	explained (cum.)	factor sd per day	level loadings	slope loadings
CL	2026-05-20..2026-09-16	81	{1,2,3,6,12}	96.3% / 99.5% / 100.0%	4.442, 0.812, 0.299	+0.60 +0.53 +0.47 +0.32 -0.51	-0.11 +0.22 +0.57 +0.56
CL	1990-01-01:1999-12-31	2175	EIA 1-4	95.5% / 99.4% / 99.9%	0.944, 0.189, 0.073	+0.61 +0.51 +0.45 +0.41	-0.79 +0.32 +0.38 +0.37
CL	2000-01-01:2009-12-31	2271	EIA 1-4	97.7% / 99.8% / 100.0%	2.809, 0.405, 0.132	+0.54 +0.50 +0.49 +0.47	-0.83 +0.17 +0.32 +0.43
CL	2010-01-01:2019-12-31	2514	EIA 1-4	99.3% / 99.9% / 100.0%	2.596, 0.206, 0.061	+0.51 +0.51 +0.50 +0.49	-0.77 -0.06 +0.33 +0.54
CL	2020-01-01:2024-12-31	1068	EIA 1-4	99.1% / 99.8% / 100.0%	3.785, 0.306, 0.181	+0.53 +0.52 +0.49 +0.46	-0.78 +0.01 +0.31 +0.55
CL	2010-01-01:2024-12-31	3583	EIA 1-4	99.2% / 99.8% / 100.0%	2.999, 0.243, 0.114	+0.52 +0.51 +0.49 +0.47	-0.77 -0.04 +0.32 +0.56
NG	2026-05-27..2026-09-17	78	{1,2,3,6,12}	57.4% / 93.8% / 99.3%	0.132, 0.105, 0.041	+0.46 +0.51 +0.68 -0.24 -0.18	+0.19 +0.04 -0.92 -0.16

NG	1990-01-01:1999-12-31	1314	EIA 1-4	88.5% / 97.5% / 99.4%	0.144, 0.046, 0.021	+0.68 +0.53 +0.41 +0.36	0.62 +0.06 +0.50 +0.60
NG	2000-01-01:2009-12-31	2246	EIA 1-4	90.6% / 97.1% / 99.2%	0.444, 0.118, 0.067	+0.55 +0.54 +0.49 +0.41	0.68 -0.13 +0.37 +0.62
NG	2010-01-01:2019-12-31	2517	EIA 1-4	92.5% / 97.8% / 99.4%	0.170, 0.041, 0.022	+0.55 +0.53 +0.48 +0.44	0.60 -0.21 +0.25 +0.73
NG	2020-01-01:2024-12-31	1068	EIA 1-4	92.8% / 98.7% / 99.9%	0.376, 0.095, 0.041	+0.55 +0.50 +0.49 +0.46	0.79 +0.07 +0.34 +0.50
NG	2010-01-01:2024-12-31	3586	EIA 1-4	92.7% / 98.3% / 99.7%	0.250, 0.061, 0.031	+0.55 +0.51 +0.48 +0.45	0.75 -0.01 +0.33 +0.57

WTI crude (CL): price impact from hourly bars (data/derived/bars_cl_1h.csv)

Pooled self-impact sign(Q) R/sigma = Y |Q|^delta on windows of h hourly bars, Q = tick-rule signed volume / daily volume; delta from a log-log fit on 16 volume bins, 95% CI by block bootstrap over days. Cross-impact: ridge of each contract's return on sign(Q_i)|Q_i|^delta of all contracts; R^2 on the raw windows. Decay: cumulative front-contract response after the window relative to the concurrent response.

window										
1h	1426	76	12	0.206 [0.174, 0.241]	1.298	0.975 / 0.043	-4.359 / -0.253	0.357	0.428	1.01 / 1.02 / 1.06 / 1
3h	475	76	12	0.371 [0.302, 0.462]	1.360	0.907 / 0.056	-0.640 / -0.087	0.264	0.384	1.04 / 0.90 / 1.07 / 1
6h	237	76	12	0.413 [0.300, 0.716]	1.034	0.855 / 0.051	0.137 / -0.022	0.194	0.316	1.09 / 1.31 / 1.37 / 1

Simulator model on the {1,2,3,6,12}-month ladder: delta 0.206, sigma per window 0.5908, 0.5391, 0.4741, 0.3917, 0.2979, daily volume 137925, 83450, 84671, 23354, 4854 contracts, Y 1.298, 1.298, 1.298, 1.298, 1.298, transient decay 0.13 per window, permanent fraction 1.00.

Henry Hub gas (NG): price impact from hourly bars (data/derived/bars_ng_1h.csv)

Pooled self-impact sign(Q) R/sigma = Y |Q|^delta on windows of h hourly bars, Q = tick-rule signed volume / daily volume; delta from a log-log fit on 16 volume bins, 95% CI by block bootstrap over days. Cross-impact: ridge of each contract's return on sign(Q_i)|Q_i|^delta of all contracts; R^2 on the raw windows. Decay: cumulative front-contract response after the window relative to the concurrent response.

window										
1h	1428	76	12	0.216 [0.196, 0.241]	1.516	0.982 / 0.100	-1.461 / -0.209	0.451	0.517	0.94 / 0.89 / 0.97 / 0
3h	473	76	12	0.354 [0.312, 0.411]	1.480	0.892 / 0.158	0.496 / 0.070	0.365	0.486	0.98 / 0.91 / 0.62 / 0
6h	235	76	12	0.430 [0.309, 0.590]	1.322	0.907 / 0.141	0.640 / 0.090	0.320	0.435	0.82 / 0.66 / 0.68 / 0

Simulator model on the {1,2,3,6,12}-month ladder: delta 0.216, sigma per window 0.0144, 0.0127, 0.0098, 0.0085, 0.0052, daily volume 76430, 42764, 22639, 15040, 2914 contracts, Y 1.516, 1.516, 1.516, 1.516, 1.516, transient decay 0.05 per window, permanent fraction 0.90.

WTI crude (CL): hedging policies on 400 random swap books (gamma 5e-07, horizon 0.25 days)

Expected execution cost (spread + transient + permanent impact, always under the full concave cross-impact model) per $1M of book notional, with the residual risk of the hedged book; 95% bootstrap CIs over books. Rows marked @FrontRisk are the impact-aware hedge and its ablations dialled (log-bisection on gamma) to the front stack's residual risk, so the cost saving is at equal risk.

policy							
NoHedge		0	[0, 0]	291056	-100.0%	[-100.0, -100.0]	0
FrontStack		1961	[1897, 2026]	100265	+0.0%	[+0.0, +0.0]	136
Strip		2075	[2048, 2101]	0	+210.5%	[+32.3, +442.8]	158
FactorMin		1915	[1861, 1970]	8640	+136.0%	[+4.0, +320.3]	143
ImpactAware		361	[331, 390]	194068	-84.3%	[-85.5, -83.2]	42
ImpactAware_noCross@FS	yes	996	[943, 1046]	97679	-50.3%	[-52.9, -47.8]	80
ImpactAware_linear@FS	yes	955	[904, 1003]	97845	-52.1%	[-54.7, -49.6]	75
ImpactAware@FrontRisk	yes	918	[868, 967]	97760	-54.3%	[-56.7, -51.9]	82

Henry Hub gas (NG): hedging policies on 400 random swap books (gamma 5e-07, horizon 0.25 days)

Expected execution cost (spread + transient + permanent impact, always under the full concave cross-impact model) per $1M of book notional, with the residual risk of the hedged book; 95% bootstrap CIs over books. Rows marked @FrontRisk are the impact-aware hedge and its ablations dialled (log-bisection on gamma) to the front stack's residual risk, so the cost saving is at equal risk.

policy							
NoHedge		0	[0, 0]	85774	-100.0%	[-100.0, -100.0]	0
FrontStack		1557	[1506, 1607]	40130	+0.0%	[+0.0, +0.0]	136
Strip		1733	[1714, 1752]	0	+203.2%	[+38.5, +415.0]	158
FactorMin		1363	[1322, 1404]	21265	+38.1%	[-6.7, +100.5]	125
ImpactAware		2	[1, 3]	85328	-99.9%	[-99.9, -99.9]	1
ImpactAware_noCross@FS	yes	659	[617, 699]	39066	-59.9%	[-62.2, -57.8]	69
ImpactAware_linear@FS	yes	647	[607, 686]	39137	-60.3%	[-62.5, -58.1]	68

ImpactAware@FrontRisk	yes	624	[584, 663]	38969	-62.1%	[-64.2, -60.1]	71	

WTI crude (CL): RFQ simulator, 40 synthetic days x 8 seeds (common random numbers)

P&L per day with 95% bootstrap CI over seeds; decomposition spread + inventory + adverse + hedge execution + hedge impact = total (residual is machine zero); 5-minute P&L std; daily P&L per unit of 5-minute variance; internalisation = 1 - hedged / client contracts; hit ratios consumer / producer / fund.

														hedger
FrontStack	Symmetric	2e-07	50.0	[26.4, 72.7]	90.1	-14.5	-4.3	-11.6	-9.7	13.51	0.29	0.67	505	0.42/0.44/0.410
FrontStack	BEGV	2e-07	57.5	[38.6, 76.1]	89.1	-14.2	-3.9	-7.3	-6.2	12.26	0.40	0.77	346	0.43/0.44/0.330
FrontStack	BEGV_adv	2e-07	58.7	[39.2, 80.4]	86.9	-13.1	-3.1	-6.2	-5.8	12.05	0.43	0.78	324	0.44/0.44/0.250
FrontStack	BEGV_adv_ca	2e-07	62.5	[44.4, 82.5]	87.0	-9.4	-3.0	-6.2	-5.9	12.05	0.45	0.78	327	0.44/0.44/0.250
FrontStack	Oracle	2e-07	66.4	[48.4, 84.3]	86.2	-4.4	-2.9	-6.2	-6.3	12.03	0.48	0.78	332	0.44/0.44/0.240
Strip	Symmetric	2e-07	37.4	[10.2, 63.6]	85.9	-13.8	-5.3	-15.4	-14.1	12.23	0.27	0.54	702	0.42/0.44/0.410
Strip	BEGV	2e-07	47.9	[26.8, 69.8]	89.2	-13.0	-4.4	-11.7	-12.1	11.40	0.41	0.63	557	0.44/0.44/0.320
Strip	BEGV_adv	2e-07	50.4	[28.4, 73.2]	87.9	-13.0	-3.3	-10.1	-11.0	11.26	0.45	0.65	530	0.44/0.44/0.260
Strip	BEGV_adv_ca	2e-07	54.2	[33.9, 75.8]	87.8	-9.1	-3.4	-10.1	-11.0	11.24	0.47	0.65	533	0.45/0.44/0.250
Strip	Oracle	2e-07	59.6	[39.5, 80.5]	86.9	-2.8	-3.4	-10.1	-11.1	11.27	0.51	0.64	539	0.45/0.44/0.240
ImpactAware	Symmetric	2e-07	46.7	[22.4, 68.9]	90.1	-17.4	-4.5	-11.5	-10.0	13.40	0.28	0.67	501	0.42/0.44/0.410
ImpactAware	BEGV	2e-07	56.9	[35.6, 77.4]	89.1	-14.7	-4.0	-7.2	-6.3	12.21	0.40	0.77	341	0.44/0.44/0.320
ImpactAware	BEGV_adv	2e-07	57.7	[38.0, 77.3]	87.4	-13.9	-3.3	-6.3	-6.2	12.12	0.42	0.78	329	0.44/0.44/0.260
ImpactAware	BEGV_adv_ca	2e-07	61.4	[42.6, 80.0]	87.1	-10.2	-3.2	-6.2	-6.2	12.02	0.44	0.78	328	0.44/0.44/0.250
ImpactAware	Oracle	2e-07	64.4	[45.7, 81.5]	86.4	-6.3	-3.1	-6.2	-6.4	12.04	0.46	0.77	334	0.44/0.44/0.250
FrontStack	Symmetric	5e-07	50.0	[26.4, 72.7]	90.1	-14.5	-4.3	-11.6	-9.7	13.51	0.29	0.67	505	0.42/0.44/0.410
FrontStack	BEGV	5e-07	62.1	[41.9, 82.2]	87.4	-11.1	-3.8	-5.5	-4.8	11.78	0.48	0.82	271	0.43/0.44/0.310
FrontStack	BEGV_adv	5e-07	59.5	[38.7, 80.4]	85.6	-13.6	-3.0	-4.9	-4.6	11.72	0.48	0.82	265	0.43/0.44/0.240
FrontStack	BEGV_adv_ca	5e-07	66.2	[49.0, 83.9]	85.4	-6.4	-3.1	-4.9	-4.7	11.79	0.50	0.82	268	0.43/0.44/0.240
FrontStack	Oracle	5e-07	68.9	[50.3, 87.9]	85.0	-3.7	-3.1	-5.0	-4.4	11.81	0.52	0.82	274	0.43/0.44/0.230
Strip	Symmetric	5e-07	37.4	[10.2, 63.6]	85.9	-13.8	-5.3	-15.4	-14.1	12.23	0.27	0.54	702	0.42/0.44/0.410
Strip	BEGV	5e-07	50.4	[29.0, 72.2]	87.7	-12.3	-4.2	-9.8	-11.0	11.05	0.46	0.67	484	0.43/0.44/0.310
Strip	BEGV_adv	5e-07	52.1	[30.9, 74.6]	86.3	-12.3	-3.3	-8.6	-10.0	10.93	0.49	0.69	464	0.43/0.44/0.240
Strip	BEGV_adv_ca	5e-07	55.9	[36.0, 77.1]	86.0	-8.5	-3.3	-8.4	-9.9	10.92	0.52	0.69	461	0.43/0.44/0.230
Strip	Oracle	5e-07	61.0	[41.2, 81.2]	85.1	-2.6	-3.2	-8.5	-9.8	11.03	0.55	0.68	471	0.44/0.43/0.230
ImpactAware	Symmetric	5e-07	46.7	[22.4, 68.9]	90.1	-17.4	-4.5	-11.5	-10.0	13.40	0.28	0.67	501	0.42/0.44/0.410
ImpactAware	BEGV	5e-07	58.6	[37.2, 80.0]	87.7	-14.4	-3.8	-5.5	-5.3	11.83	0.45	0.81	271	0.43/0.44/0.310
ImpactAware	BEGV_adv	5e-07	60.5	[39.6, 81.1]	86.1	-12.4	-3.2	-5.0	-5.0	11.76	0.47	0.81	272	0.43/0.45/0.240
ImpactAware	BEGV_adv_ca	5e-07	64.8	[45.7, 84.7]	85.8	-8.0	-3.2	-4.9	-4.8	11.78	0.49	0.82	269	0.43/0.44/0.230
ImpactAware	Oracle	5e-07	67.3	[48.0, 86.4]	84.9	-4.8	-3.1	-4.9	-4.9	11.84	0.51	0.82	269	0.43/0.44/0.230
FrontStack	Symmetric	1e-06	50.0	[26.4, 72.7]	90.1	-14.5	-4.3	-11.6	-9.7	13.51	0.29	0.67	505	0.42/0.44/0.410
FrontStack	BEGV	1e-06	65.2	[46.4, 83.7]	85.5	-8.0	-3.7	-4.4	-4.2	11.39	0.54	0.85	223	0.43/0.44/0.230
FrontStack	BEGV_adv	1e-06	62.3	[42.1, 83.0]	84.2	-11.5	-2.9	-3.8	-3.7	11.35	0.53	0.86	208	0.43/0.43/0.220
FrontStack	BEGV_adv_ca	1e-06	64.7	[46.4, 83.1]	84.7	-9.4	-3.1	-3.8	-3.7	11.32	0.56	0.86	212	0.43/0.44/0.220
FrontStack	Oracle	1e-06	66.6	[49.9, 83.6]	83.7	-6.8	-2.9	-3.9	-3.5	11.45	0.54	0.85	220	0.42/0.43/0.210
Strip	Symmetric	1e-06	37.4	[10.2, 63.6]	85.9	-13.8	-5.3	-15.4	-14.1	12.23	0.27	0.54	702	0.42/0.44/0.410
Strip	BEGV	1e-06	54.6	[33.7, 75.7]	85.8	-9.7	-3.9	-8.2	-9.4	10.67	0.54	0.71	421	0.43/0.44/0.290
Strip	BEGV_adv	1e-06	54.4	[33.6, 76.4]	84.7	-11.1	-3.2	-7.3	-8.7	10.55	0.55	0.72	412	0.42/0.43/0.230
Strip	BEGV_adv_ca	1e-06	57.7	[38.4, 78.1]	84.7	-7.8	-3.2	-7.3	-8.7	10.50	0.59	0.72	413	0.42/0.43/0.230
Strip	Oracle	1e-06	63.4	[44.5, 82.5]	84.0	-1.3	-3.1	-7.4	-8.7	10.69	0.60	0.71	424	0.42/0.43/0.220
ImpactAware	Symmetric	1e-06	49.3	[23.5, 72.8]	90.1	-15.0	-4.5	-11.4	-9.9	13.31	0.30	0.67	498	0.42/0.44/0.410
ImpactAware	BEGV	1e-06	60.6	[39.5, 81.8]	85.5	-12.5	-3.6	-4.2	-4.5	11.33	0.50	0.85	217	0.43/0.44/0.300
ImpactAware	BEGV_adv	1e-06	61.2	[41.0, 81.4]	84.5	-12.8	-3.1	-3.7	-3.7	11.34	0.53	0.86	205	0.42/0.43/0.220
ImpactAware	BEGV_adv_ca	1e-06	62.1	[43.4, 81.2]	84.4	-11.4	-3.2	-3.8	-3.9	11.35	0.53	0.85	213	0.42/0.44/0.220
ImpactAware	Oracle	1e-06	69.0	[49.4, 87.6]	83.5	-3.9	-3.1	-3.8	-3.7	11.41	0.57	0.85	215	0.42/0.43/0.220
FrontStack	Symmetric	2e-06	50.0	[26.4, 72.7]	90.1	-14.5	-4.3	-11.6	-9.7	13.51	0.29	0.67	505	0.42/0.44/0.410
FrontStack	BEGV	2e-06	62.7	[42.8, 82.5]	82.7	-10.5	-3.4	-3.0	-3.1	10.79	0.58	0.89	157	0.41/0.43/0.270
FrontStack	BEGV_adv	2e-06	63.8	[45.2, 82.9]	82.5	-10.4	-2.8	-2.8	-2.7	10.72	0.62	0.89	157	0.41/0.43/0.210
FrontStack	BEGV_adv_ca	2e-06	66.0	[48.1, 83.8]	82.3	-8.0	-2.8	-2.8	-2.7	10.73	0.64	0.89	160	0.42/0.44/0.210
FrontStack	Oracle	2e-06	72.0	[55.6, 88.0]	81.4	-1.2	-2.8	-2.8	-2.6	10.94	0.64	0.89	162	0.42/0.43/0.210
Strip	Symmetric	2e-06	37.4	[10.2, 63.6]	85.9	-13.8	-5.3	-15.4	-14.1	12.23	0.27	0.54	702	0.42/0.44/0.410

Strip	BEGV	2e-06	58.8	[40.8, 76.8]	83.8	-7.5	-3.4	-6.5	-7.7	10.14	0.65	0.76	349	0.42/0.43/0.270
Strip	BEGV_adv	2e-06	58.2	[41.1, 76.2]	83.1	-8.9	-3.0	-6.0	-7.0	9.89	0.67	0.77	348	0.42/0.43/0.220
Strip	BEGV_adv_ca	2e-06	61.2	[45.9, 77.1]	82.8	-5.4	-3.1	-6.0	-7.2	9.90	0.70	0.76	350	0.42/0.43/0.220
Strip	Oracle	2e-06	66.8	[52.9, 81.5]	81.8	1.2	-3.0	-6.0	-7.1	10.10	0.71	0.76	358	0.42/0.43/0.210
ImpactAware	Symmetric	2e-06	47.8	[22.9, 70.0]	90.1	-15.7	-4.5	-11.5	-10.7	13.40	0.29	0.67	499	0.42/0.44/0.410
ImpactAware	BEGV	2e-06	64.1	[46.6, 82.4]	83.4	-9.3	-3.4	-3.1	-3.4	10.72	0.61	0.89	164	0.42/0.44/0.230
ImpactAware	BEGV_adv	2e-06	67.0	[46.9, 87.8]	82.8	-7.0	-2.9	-2.8	-3.1	10.62	0.64	0.89	159	0.42/0.44/0.220
ImpactAware	BEGV_adv_ca	2e-06	68.8	[50.9, 86.9]	82.4	-4.7	-2.9	-2.8	-3.2	10.65	0.67	0.89	159	0.42/0.43/0.220
ImpactAware	Oracle	2e-06	73.3	[56.3, 90.2]	81.5	0.4	-3.0	-2.7	-3.0	10.90	0.67	0.89	158	0.42/0.43/0.210
FrontStack	Symmetric	5e-06	50.0	[26.4, 72.7]	90.1	-14.5	-4.3	-11.6	-9.7	13.51	0.29	0.67	505	0.42/0.44/0.410
FrontStack	BEGV	5e-06	64.0	[47.2, 80.1]	78.1	-7.7	-3.1	-1.6	-1.7	9.39	0.75	0.94	92	0.39/0.43/0.250
FrontStack	BEGV_adv	5e-06	65.8	[50.0, 81.2]	76.7	-5.5	-2.4	-1.4	-1.5	9.55	0.80	0.94	86	0.38/0.42/0.200
FrontStack	BEGV_adv_ca	5e-06	69.0	[55.3, 82.6]	76.7	-2.4	-2.5	-1.4	-1.5	9.22	0.85	0.94	86	0.38/0.42/0.200
FrontStack	Oracle	5e-06	71.7	[57.3, 85.7]	76.2	0.8	-2.5	-1.4	-1.5	9.84	0.80	0.94	87	0.37/0.43/0.190
Strip	Symmetric	5e-06	37.4	[10.2, 63.6]	85.9	-13.8	-5.3	-15.4	-14.1	12.23	0.27	0.54	702	0.42/0.44/0.410
Strip	BEGV	5e-06	58.2	[41.3, 73.8]	79.0	-7.7	-3.1	-4.4	-5.6	9.40	0.74	0.83	256	0.39/0.42/0.250
Strip	BEGV_adv	5e-06	59.5	[45.0, 74.3]	78.5	-7.1	-2.8	-4.1	-5.1	9.13	0.79	0.83	251	0.40/0.42/0.200
Strip	BEGV_adv_ca	5e-06	63.4	[49.4, 77.6]	78.2	-3.1	-2.7	-3.9	-5.1	9.17	0.83	0.83	246	0.40/0.42/0.200
Strip	Oracle	5e-06	67.8	[53.6, 81.0]	77.2	2.7	-2.5	-4.1	-5.5	9.45	0.81	0.82	265	0.40/0.42/0.190
ImpactAware	Symmetric	5e-06	48.3	[22.5, 72.1]	90.1	-14.0	-4.7	-11.0	-12.1	13.31	0.30	0.68	479	0.42/0.44/0.410
ImpactAware	BEGV	5e-06	62.5	[44.2, 80.2]	78.0	-8.5	-2.9	-1.6	-2.4	9.73	0.72	0.93	93	0.38/0.43/0.250
ImpactAware	BEGV_adv	5e-06	67.6	[52.1, 82.7]	77.7	-4.1	-2.5	-1.4	-2.1	9.55	0.82	0.94	85	0.38/0.43/0.220
ImpactAware	BEGV_adv_ca	5e-06	68.6	[53.9, 82.9]	77.4	-2.9	-2.5	-1.4	-2.1	9.63	0.82	0.94	86	0.39/0.43/0.210
ImpactAware	Oracle	5e-06	72.0	[56.0, 87.0]	76.7	1.6	-2.6	-1.4	-2.2	9.74	0.80	0.94	89	0.39/0.43/0.210

Simulator: each quoter minus Symmetric on the same seeds (paired bootstrap 95% CI over seeds):

						hedger
FrontStack	2e-07	BEGV	+7.4	[+1.3, +13.1]	+14.9%	+0.11
FrontStack	2e-07	BEGV_adv	+8.6	[-1.1, +17.2]	+17.3%	+0.14
FrontStack	2e-07	BEGV_adv_cal	+12.5	[+2.5, +21.9]	+24.9%	+0.16
FrontStack	2e-07	Oracle	+16.4	[+7.6, +25.0]	+32.7%	+0.19
FrontStack	5e-07	BEGV	+12.1	[+4.1, +19.7]	+24.1%	+0.19
FrontStack	5e-07	BEGV_adv	+9.4	[+1.0, +16.9]	+18.8%	+0.19
FrontStack	5e-07	BEGV_adv_cal	+16.2	[+6.7, +26.6]	+32.4%	+0.21
FrontStack	5e-07	Oracle	+18.8	[+9.6, +28.1]	+37.7%	+0.23
FrontStack	1e-06	BEGV	+15.2	[+6.7, +24.0]	+30.4%	+0.25
FrontStack	1e-06	BEGV_adv	+12.3	[+4.8, +19.5]	+24.5%	+0.24
FrontStack	1e-06	BEGV_adv_cal	+14.6	[+5.7, +23.0]	+29.3%	+0.27
FrontStack	1e-06	Oracle	+16.6	[+7.2, +26.2]	+33.2%	+0.25
FrontStack	2e-06	BEGV	+12.7	[+3.2, +22.2]	+25.3%	+0.29
FrontStack	2e-06	BEGV_adv	+13.8	[+2.4, +26.2]	+27.6%	+0.33
FrontStack	2e-06	BEGV_adv_cal	+16.0	[+4.8, +27.6]	+32.0%	+0.35
FrontStack	2e-06	Oracle	+22.0	[+10.8, +33.5]	+44.0%	+0.35
FrontStack	5e-06	BEGV	+14.0	[+1.9, +27.2]	+27.9%	+0.46
FrontStack	5e-06	BEGV_adv	+15.8	[+7.0, +25.3]	+31.6%	+0.51
FrontStack	5e-06	BEGV_adv_cal	+18.9	[+8.8, +29.7]	+37.8%	+0.56
FrontStack	5e-06	Oracle	+21.7	[+10.2, +33.6]	+43.3%	+0.51
Strip	2e-07	BEGV	+10.5	[+3.0, +18.4]	+28.0%	+0.14
Strip	2e-07	BEGV_adv	+13.0	[+4.2, +22.1]	+34.7%	+0.18
Strip	2e-07	BEGV_adv_cal	+16.7	[+7.4, +27.4]	+44.7%	+0.20
Strip	2e-07	Oracle	+22.2	[+13.1, +31.4]	+59.2%	+0.24
Strip	5e-07	BEGV	+12.9	[+5.4, +21.0]	+34.6%	+0.19
Strip	5e-07	BEGV_adv	+14.7	[+6.0, +23.5]	+39.2%	+0.22
Strip	5e-07	BEGV_adv_cal	+18.5	[+9.3, +29.4]	+49.4%	+0.25
Strip	5e-07	Oracle	+23.6	[+14.6, +33.3]	+63.0%	+0.28
Strip	1e-06	BEGV	+17.1	[+10.0, +25.2]	+45.8%	+0.27
Strip	1e-06	BEGV_adv	+17.0	[+7.1, +26.6]	+45.3%	+0.28
Strip	1e-06	BEGV_adv_cal	+20.3	[+9.8, +32.4]	+54.1%	+0.32
Strip	1e-06	Oracle	+26.0	[+16.0, +36.8]	+69.5%	+0.33

Strip	2e-06	BEGV	+21.4	[+11.4, +31.9]	+57.1%	+0.38
Strip	2e-06	BEGV_adv	+20.8	[+10.2, +33.0]	+55.6%	+0.39
Strip	2e-06	BEGV_adv_cal	+23.7	[+11.7, +37.8]	+63.4%	+0.43
Strip	2e-06	Oracle	+29.4	[+16.0, +43.8]	+78.6%	+0.43
Strip	5e-06	BEGV	+20.8	[+9.1, +33.7]	+55.5%	+0.47
Strip	5e-06	BEGV_adv	+22.1	[+9.1, +35.7]	+59.1%	+0.52
Strip	5e-06	BEGV_adv_cal	+26.0	[+13.0, +40.2]	+69.4%	+0.56
Strip	5e-06	Oracle	+30.4	[+16.2, +45.0]	+81.1%	+0.54
ImpactAware	2e-07	BEGV	+10.2	[+5.2, +15.2]	+21.9%	+0.12
ImpactAware	2e-07	BEGV_adv	+11.0	[+4.0, +17.6]	+23.6%	+0.14
ImpactAware	2e-07	BEGV_adv_cal	+14.7	[+7.4, +22.2]	+31.5%	+0.16
ImpactAware	2e-07	Oracle	+17.7	[+9.6, +25.8]	+37.9%	+0.18
ImpactAware	5e-07	BEGV	+11.9	[+4.8, +18.1]	+25.5%	+0.17
ImpactAware	5e-07	BEGV_adv	+13.8	[+6.3, +20.6]	+29.5%	+0.19
ImpactAware	5e-07	BEGV_adv_cal	+18.1	[+8.2, +27.1]	+38.7%	+0.21
ImpactAware	5e-07	Oracle	+20.6	[+8.7, +32.0]	+44.1%	+0.23
ImpactAware	1e-06	BEGV	+11.3	[+1.0, +20.4]	+23.0%	+0.20
ImpactAware	1e-06	BEGV_adv	+12.0	[+1.4, +21.1]	+24.4%	+0.23
ImpactAware	1e-06	BEGV_adv_cal	+12.8	[+0.4, +23.3]	+26.0%	+0.23
ImpactAware	1e-06	Oracle	+19.7	[+8.1, +30.1]	+40.1%	+0.27
ImpactAware	2e-06	BEGV	+16.4	[+5.2, +26.5]	+34.3%	+0.32
ImpactAware	2e-06	BEGV_adv	+19.2	[+9.6, +29.2]	+40.3%	+0.35
ImpactAware	2e-06	BEGV_adv_cal	+21.0	[+11.6, +30.5]	+44.0%	+0.38
ImpactAware	2e-06	Oracle	+25.5	[+13.6, +37.8]	+53.5%	+0.38
ImpactAware	5e-06	BEGV	+14.2	[+4.7, +24.6]	+29.4%	+0.43
ImpactAware	5e-06	BEGV_adv	+19.2	[+9.3, +30.0]	+39.8%	+0.52
ImpactAware	5e-06	BEGV_adv_cal	+20.2	[+9.2, +32.6]	+41.9%	+0.52
ImpactAware	5e-06	Oracle	+23.7	[+12.5, +36.0]	+48.9%	+0.51

Simulator: each hedger minus FrontStack on the same seeds:

						quoter
BEGV	2e-07	Strip	-9.6	[-17.1, -2.4]	-10.3	11.40 vs 12.26
BEGV	2e-07	ImpactAware	-0.5	[-4.1, +2.7]	+0.0	12.21 vs 12.26
BEGV	5e-07	Strip	-11.7	[-16.6, -6.3]	-10.4	11.05 vs 11.78
BEGV	5e-07	ImpactAware	-3.5	[-7.3, -0.2]	-0.5	11.83 vs 11.78
BEGV	1e-06	Strip	-10.7	[-16.2, -4.0]	-9.1	10.67 vs 11.39
BEGV	1e-06	ImpactAware	-4.7	[-9.4, -0.4]	-0.2	11.33 vs 11.39
BEGV	2e-06	Strip	-3.9	[-9.5, +0.8]	-8.1	10.14 vs 10.79
BEGV	2e-06	ImpactAware	+1.5	[-2.6, +5.2]	-0.4	10.72 vs 10.79
BEGV	5e-06	Strip	-5.8	[-10.8, -1.1]	-6.8	9.40 vs 9.39
BEGV	5e-06	ImpactAware	-1.5	[-4.4, +1.4]	-0.7	9.73 vs 9.39
BEGV_adv	2e-07	Strip	-8.3	[-19.5, +2.8]	-9.1	11.26 vs 12.05
BEGV_adv	2e-07	ImpactAware	-1.0	[-8.2, +5.4]	-0.5	12.12 vs 12.05
BEGV_adv	5e-07	Strip	-7.3	[-15.9, +0.5]	-9.0	10.93 vs 11.72
BEGV_adv	5e-07	ImpactAware	+1.0	[-3.2, +4.9]	-0.5	11.76 vs 11.72
BEGV_adv	1e-06	Strip	-7.9	[-15.7, -0.1]	-8.6	10.55 vs 11.35
BEGV_adv	1e-06	ImpactAware	-1.0	[-3.9, +1.5]	+0.0	11.34 vs 11.35
BEGV_adv	2e-06	Strip	-5.6	[-9.0, -2.4]	-7.6	9.89 vs 10.72
BEGV_adv	2e-06	ImpactAware	+3.2	[-1.5, +8.7]	-0.4	10.62 vs 10.72
BEGV_adv	5e-06	Strip	-6.3	[-12.3, -0.6]	-6.2	9.13 vs 9.55
BEGV_adv	5e-06	ImpactAware	+1.7	[-2.0, +4.7]	-0.6	9.55 vs 9.55
BEGV_adv_cal	2e-07	Strip	-8.3	[-17.9, +1.2]	-9.0	11.24 vs 12.05
BEGV_adv_cal	2e-07	ImpactAware	-1.1	[-8.0, +5.1]	-0.2	12.02 vs 12.05
BEGV_adv_cal	5e-07	Strip	-10.3	[-17.7, -2.9]	-8.6	10.92 vs 11.79
BEGV_adv_cal	5e-07	ImpactAware	-1.5	[-5.7, +2.8]	-0.1	11.78 vs 11.79
BEGV_adv_cal	1e-06	Strip	-7.0	[-15.2, +0.7]	-8.5	10.50 vs 11.32
BEGV_adv_cal	1e-06	ImpactAware	-2.6	[-5.4, -0.0]	-0.2	11.35 vs 11.32
BEGV_adv_cal	2e-06	Strip	-4.9	[-8.9, -0.7]	-7.7	9.90 vs 10.73
BEGV_adv_cal	2e-06	ImpactAware	+2.7	[-1.1, +6.6]	-0.5	10.65 vs 10.73

BEGV_adv_cal	5e-06	Strip	-5.5	[-10.2, -0.8]	-6.2	9.17 vs 9.22
BEGV_adv_cal	5e-06	ImpactAware	-0.4	[-3.4, +2.7]	-0.6	9.63 vs 9.22
Oracle	2e-07	Strip	-6.8	[-15.5, +2.6]	-8.6	11.27 vs 12.03
Oracle	2e-07	ImpactAware	-2.0	[-7.7, +3.2]	-0.2	12.04 vs 12.03
Oracle	5e-07	Strip	-7.9	[-12.8, -2.3]	-8.9	11.03 vs 11.81
Oracle	5e-07	ImpactAware	-1.6	[-4.9, +2.0]	-0.4	11.84 vs 11.81
Oracle	1e-06	Strip	-3.2	[-9.7, +3.5]	-8.7	10.69 vs 11.45
Oracle	1e-06	ImpactAware	+2.4	[-0.3, +5.2]	-0.1	11.41 vs 11.45
Oracle	2e-06	Strip	-5.2	[-9.0, -1.9]	-7.8	10.10 vs 10.94
Oracle	2e-06	ImpactAware	+1.2	[-1.7, +4.2]	-0.4	10.90 vs 10.94
Oracle	5e-06	Strip	-3.9	[-8.7, +0.9]	-6.8	9.45 vs 9.84
Oracle	5e-06	ImpactAware	+0.3	[-4.6, +4.9]	-0.8	9.74 vs 9.84
Symmetric	2e-07	Strip	-12.6	[-23.1, -0.6]	-8.1	12.23 vs 13.51
Symmetric	2e-07	ImpactAware	-3.3	[-8.3, +1.1]	-0.2	13.40 vs 13.51
Symmetric	5e-07	Strip	-12.6	[-23.1, -0.6]	-8.1	12.23 vs 13.51
Symmetric	5e-07	ImpactAware	-3.3	[-8.3, +1.1]	-0.2	13.40 vs 13.51
Symmetric	1e-06	Strip	-12.6	[-23.1, -0.6]	-8.1	12.23 vs 13.51
Symmetric	1e-06	ImpactAware	-0.8	[-5.8, +4.0]	-0.0	13.31 vs 13.51
Symmetric	2e-06	Strip	-12.6	[-23.1, -0.6]	-8.1	12.23 vs 13.51
Symmetric	2e-06	ImpactAware	-2.3	[-7.1, +2.4]	-0.9	13.40 vs 13.51
Symmetric	5e-06	Strip	-12.6	[-23.1, -0.6]	-8.1	12.23 vs 13.51
Symmetric	5e-06	ImpactAware	-1.7	[-7.6, +3.8]	-1.8	13.31 vs 13.51

Mark-outs at 30 minutes against the dealer ($/bbl per contract, positive = the client was right), gamma 1e-06, hedger ImpactAware:

							quoter
Symmetric	consumer	normal	191	0.3811	-0.0043	+0.0359	+0.0739
Symmetric	consumer	pre-release	2	0.3811	+0.8138	+0.3853	-0.9455
Symmetric	producer	normal	173	0.4264	+0.0109	+0.0014	+0.0368
Symmetric	producer	pre-release	3	0.4264	-0.0243	+0.8036	+1.0379
Symmetric	fund	normal	235	0.1405	-0.0032	-0.0202	-0.0156
Symmetric	fund	pre-release	7	0.1405	-0.1480	+1.9505	+2.8581
BEGV	consumer	normal	193	0.3767	-0.0064	+0.0386	+0.0668
BEGV	consumer	pre-release	3	0.3913	+0.5439	+0.3666	-0.5433
BEGV	producer	normal	172	0.4134	+0.0107	-0.0061	+0.0194
BEGV	producer	pre-release	2	0.3882	-0.1335	+1.7778	+2.4541
BEGV	fund	normal	169	0.1689	+0.0022	-0.0199	+0.0276
BEGV	fund	pre-release	7	0.1519	-0.1480	+1.9394	+2.8460
BEGV_adv	consumer	normal	192	0.3747	-0.0059	+0.0404	+0.0685
BEGV_adv	consumer	pre-release	3	0.3827	+0.5513	+0.3739	-0.5359
BEGV_adv	producer	normal	169	0.4153	+0.0090	-0.0095	+0.0115
BEGV_adv	producer	pre-release	2	0.3863	-0.1335	+1.7778	+2.4541
BEGV_adv	fund	normal	127	0.2087	+0.0121	-0.0057	+0.0527
BEGV_adv	fund	pre-release	6	0.1868	-0.1675	+2.2171	+3.2540
BEGV_adv_cal	consumer	normal	191	0.3750	-0.0044	+0.0456	+0.0683
BEGV_adv_cal	consumer	pre-release	3	0.3932	+0.5439	+0.3657	-0.5442
BEGV_adv_cal	producer	normal	171	0.4153	+0.0098	-0.0068	+0.0250
BEGV_adv_cal	producer	pre-release	2	0.4012	-0.1335	+1.7778	+2.4541
BEGV_adv_cal	fund	normal	128	0.2102	+0.0170	-0.0076	+0.0444
BEGV_adv_cal	fund	pre-release	3	0.2228	-0.3303	+1.6275	+2.7462
Oracle	consumer	normal	189	0.3748	-0.0073	+0.0452	+0.0679
Oracle	consumer	pre-release	3	0.3808	+0.5439	+0.3799	-0.5200
Oracle	producer	normal	168	0.4130	+0.0105	-0.0009	+0.0210
Oracle	producer	pre-release	2	0.3887	-0.1335	+1.7778	+2.4541
Oracle	fund	normal	130	0.2106	+0.0135	-0.0130	+0.0313

Variant tox (40 days x 8 seeds):

											hedger
ImpactAware	Symmetric	1e-06	42.9	[14.8, 69.2]	-3.1	-14.4	-7.2	12.23	0.25	0.54	61
ImpactAware	BEGV_adv_cal	1e-06	66.1	[46.8, 86.1]	-2.9	-7.8	-2.3	11.01	0.59	0.73	61

Variant anticipate (40 days x 8 seeds):

											hedger
ImpactAware	Symmetric	1e-06	53.2	[34.6, 70.0]	-2.3	-14.9	-7.3	11.83	0.41	0.59	0
ImpactAware	BEGV_adv_cal	1e-06	68.2	[55.0, 81.9]	-2.1	-5.6	-3.3	9.98	0.72	0.79	0

Variant impact half (40 days x 8 seeds):

											hedger
FrontStack	BEGV_adv_cal	1e-06	69.1	[51.3, 87.1]	-3.1	-2.1	-1.6	11.27	0.60	0.86	0
ImpactAware	BEGV_adv_cal	1e-06	65.6	[46.7, 85.3]	-3.1	-2.1	-1.8	11.18	0.58	0.86	0

Variant impact double (40 days x 8 seeds):

											hedger
FrontStack	BEGV_adv_cal	1e-06	55.3	[37.5, 73.7]	-3.0	-7.1	-7.7	11.35	0.48	0.86	0
ImpactAware	BEGV_adv_cal	1e-06	56.6	[39.1, 75.3]	-3.1	-7.1	-7.7	11.32	0.49	0.86	0

Henry Hub gas (NG): RFQ simulator, 40 synthetic days x 8 seeds (common random numbers)

P&L per day with 95% bootstrap CI over seeds; decomposition spread + inventory + adverse + hedge execution + hedge impact = total (residual is machine zero); 5-minute P&L std; daily P&L per unit of 5-minute variance; internalisation = 1 - hedged / client contracts; hit ratios consumer / producer / fund.

															hedger
FrontStack	Symmetric	2e-07	35.4	[20.0, 53.8]	63.8	-12.6	-5.8	-5.2	-4.9	11.59	0.27	0.71	413	0.43/0.45/0.39	0
FrontStack	BEGV	2e-07	40.3	[26.7, 55.6]	63.3	-9.9	-6.1	-3.7	-3.4	10.88	0.34	0.79	286	0.47/0.44/0.36	0
FrontStack	BEGV_adv	2e-07	44.0	[29.4, 61.2]	63.3	-7.1	-5.3	-3.6	-3.4	10.97	0.37	0.78	299	0.47/0.44/0.28	0
FrontStack	BEGV_adv_ca	2e-07	45.1	[32.0, 61.9]	63.0	-5.5	-5.3	-3.5	-3.6	10.88	0.38	0.79	298	0.47/0.44/0.28	0
FrontStack	Oracle	2e-07	44.0	[29.3, 61.7]	62.9	-6.6	-5.2	-3.6	-3.6	11.01	0.37	0.78	303	0.47/0.44/0.28	0
Strip	Symmetric	2e-07	16.9	[-3.0, 39.1]	63.4	-17.6	-7.3	-6.5	-15.1	11.60	0.14	0.63	514	0.43/0.45/0.39	0
Strip	BEGV	2e-07	29.7	[14.7, 45.3]	63.5	-10.4	-7.1	-4.8	-11.5	10.84	0.26	0.74	365	0.47/0.43/0.37	0
Strip	BEGV_adv	2e-07	32.7	[18.3, 49.7]	63.3	-8.6	-5.9	-4.6	-11.4	10.90	0.28	0.72	382	0.46/0.44/0.28	0
Strip	BEGV_adv_ca	2e-07	34.1	[20.0, 51.0]	63.0	-7.2	-5.9	-4.6	-11.2	10.83	0.29	0.73	380	0.46/0.44/0.28	0
Strip	Oracle	2e-07	34.8	[20.2, 51.7]	62.8	-6.1	-5.7	-4.6	-11.4	10.91	0.30	0.72	388	0.46/0.43/0.27	0
ImpactAware	Symmetric	2e-07	37.4	[26.2, 52.7]	64.1	-8.7	-5.7	-5.1	-7.2	11.47	0.30	0.70	409	0.43/0.45/0.39	0
ImpactAware	BEGV	2e-07	41.2	[28.5, 54.5]	63.3	-7.7	-6.1	-3.5	-4.8	10.77	0.37	0.80	277	0.47/0.43/0.37	0
ImpactAware	BEGV_adv	2e-07	44.6	[30.8, 59.7]	63.2	-5.1	-4.9	-3.5	-5.1	10.89	0.38	0.79	291	0.47/0.43/0.28	0
ImpactAware	BEGV_adv_ca	2e-07	45.7	[33.2, 60.6]	62.9	-3.8	-5.0	-3.4	-5.0	10.83	0.39	0.79	289	0.46/0.43/0.28	0
ImpactAware	Oracle	2e-07	46.3	[33.3, 60.9]	62.8	-3.3	-4.8	-3.5	-5.0	10.91	0.40	0.79	296	0.47/0.43/0.28	0
FrontStack	Symmetric	5e-07	35.4	[20.0, 53.8]	63.8	-12.6	-5.8	-5.2	-4.9	11.59	0.27	0.71	413	0.43/0.45/0.39	0
FrontStack	BEGV	5e-07	43.6	[31.3, 58.0]	61.8	-7.8	-5.6	-2.6	-2.2	10.45	0.41	0.84	215	0.45/0.45/0.35	0
FrontStack	BEGV_adv	5e-07	43.5	[30.2, 59.5]	61.6	-8.1	-5.0	-2.5	-2.5	10.40	0.40	0.84	218	0.45/0.44/0.27	0
FrontStack	BEGV_adv_ca	5e-07	44.8	[32.2, 60.4]	61.3	-6.4	-5.2	-2.5	-2.4	10.40	0.42	0.84	217	0.45/0.44/0.27	0
FrontStack	Oracle	5e-07	45.2	[32.2, 61.1]	61.1	-5.8	-4.9	-2.5	-2.7	10.55	0.41	0.84	223	0.45/0.45/0.26	0
Strip	Symmetric	5e-07	16.9	[-3.0, 39.1]	63.4	-17.6	-7.3	-6.5	-15.1	11.60	0.14	0.63	514	0.43/0.45/0.39	0
Strip	BEGV	5e-07	35.0	[21.6, 49.8]	62.0	-7.8	-6.6	-3.6	-9.0	10.37	0.33	0.79	287	0.45/0.44/0.35	0
Strip	BEGV_adv	5e-07	36.6	[23.4, 52.4]	61.8	-7.2	-5.6	-3.5	-8.9	10.48	0.34	0.79	297	0.45/0.44/0.28	0
Strip	BEGV_adv_ca	5e-07	37.6	[25.0, 53.5]	61.6	-6.2	-5.6	-3.4	-8.7	10.46	0.35	0.78	299	0.45/0.44/0.28	0
Strip	Oracle	5e-07	37.9	[24.7, 54.1]	61.3	-5.6	-5.6	-3.5	-8.8	10.56	0.35	0.78	302	0.45/0.44/0.28	0
ImpactAware	Symmetric	5e-07	37.4	[26.2, 52.7]	64.1	-8.7	-5.7	-5.1	-7.2	11.47	0.30	0.70	409	0.43/0.45/0.39	0
ImpactAware	BEGV	5e-07	44.0	[31.6, 56.6]	62.0	-6.0	-5.9	-2.5	-3.7	10.34	0.42	0.85	201	0.45/0.44/0.35	0
ImpactAware	BEGV_adv	5e-07	45.1	[33.2, 58.9]	61.9	-5.8	-4.9	-2.4	-3.8	10.38	0.42	0.84	211	0.45/0.44/0.28	0
ImpactAware	BEGV_adv_ca	5e-07	47.6	[36.8, 60.9]	61.7	-3.0	-5.0	-2.4	-3.7	10.39	0.45	0.85	209	0.45/0.44/0.27	0
ImpactAware	Oracle	5e-07	46.9	[35.2, 61.1]	61.5	-3.3	-5.1	-2.4	-3.8	10.55	0.43	0.84	216	0.45/0.44/0.27	0
FrontStack	Symmetric	1e-06	35.4	[20.0, 53.8]	63.8	-12.6	-5.8	-5.2	-4.9	11.59	0.27	0.71	413	0.43/0.45/0.39	0
FrontStack	BEGV	1e-06	45.8	[33.4, 61.2]	60.0	-5.5	-5.6	-1.8	-1.3	9.86	0.48	0.89	149	0.43/0.44/0.36	0
FrontStack	BEGV_adv	1e-06	45.7	[33.3, 61.1]	59.8	-5.9	-4.9	-1.6	-1.7	9.72	0.48	0.89	147	0.43/0.44/0.26	0
FrontStack	BEGV_adv_ca	1e-06	48.3	[36.9, 63.3]	59.6	-3.3	-4.9	-1.6	-1.5	9.73	0.51	0.89	147	0.43/0.44/0.26	0
FrontStack	Oracle	1e-06	48.0	[35.6, 64.0]	59.2	-3.1	-4.6	-1.6	-1.8	9.81	0.51	0.89	152	0.43/0.44/0.26	0
Strip	Symmetric	1e-06	16.9	[-3.0, 39.1]	63.4	-17.6	-7.3	-6.5	-15.1	11.60	0.14	0.63	514	0.43/0.45/0.39	0
Strip	BEGV	1e-06	38.5	[25.8, 52.7]	59.9	-5.8	-6.4	-2.5	-6.6	9.84	0.41	0.85	212	0.43/0.44/0.36	0
Strip	BEGV_adv	1e-06	39.4	[27.6, 54.4]	60.0	-6.1	-5.6	-2.4	-6.5	9.83	0.42	0.85	214	0.43/0.44/0.28	0

Strip	BEGV_adv_cal	1e-06	42.3	[30.4, 57.0]	59.8	-3.1	-5.5	-2.4	-6.5	9.76	0.45	0.84	218	0.43/0.44/0.280
Strip	Oracle	1e-06	41.0	[28.1, 56.3]	59.6	-4.2	-5.4	-2.4	-6.6	9.92	0.43	0.84	217	0.42/0.44/0.270
ImpactAware	Symmetric	1e-06	37.5	[26.2, 52.6]	64.1	-8.5	-5.7	-5.2	-7.2	11.48	0.30	0.70	409	0.43/0.45/0.390
ImpactAware	BEGV	1e-06	44.7	[32.8, 58.4]	59.9	-5.0	-5.8	-1.6	-2.7	9.81	0.47	0.90	138	0.43/0.44/0.340
ImpactAware	BEGV_adv	1e-06	46.7	[34.9, 60.9]	59.8	-4.1	-4.8	-1.5	-2.7	9.77	0.50	0.90	139	0.43/0.45/0.280
ImpactAware	BEGV_adv_cal	1e-06	47.7	[37.0, 61.9]	59.8	-3.4	-4.6	-1.6	-2.6	9.78	0.50	0.89	142	0.43/0.45/0.280
ImpactAware	Oracle	1e-06	47.0	[34.6, 62.3]	59.6	-3.5	-4.8	-1.6	-2.7	9.89	0.49	0.89	144	0.43/0.44/0.270
FrontStack	Symmetric	2e-06	35.4	[20.0, 53.8]	63.8	-12.6	-5.8	-5.2	-4.9	11.59	0.27	0.71	413	0.43/0.45/0.390
FrontStack	BEGV	2e-06	46.6	[36.4, 58.5]	57.4	-3.2	-5.8	-0.9	-0.9	9.00	0.59	0.94	83	0.41/0.45/0.320
FrontStack	BEGV_adv	2e-06	46.3	[35.0, 60.0]	57.3	-4.2	-5.0	-0.9	-0.9	9.04	0.57	0.94	81	0.40/0.45/0.260
FrontStack	BEGV_adv_cal	2e-06	48.0	[37.5, 60.9]	57.3	-2.6	-5.1	-0.9	-0.7	9.05	0.59	0.94	83	0.41/0.45/0.260
FrontStack	Oracle	2e-06	47.6	[35.3, 61.8]	57.2	-2.8	-5.1	-0.9	-0.8	9.21	0.57	0.94	87	0.41/0.45/0.260
Strip	Symmetric	2e-06	16.9	[-3.0, 39.1]	63.4	-17.6	-7.3	-6.5	-15.1	11.60	0.14	0.63	514	0.43/0.45/0.390
Strip	BEGV	2e-06	43.4	[32.7, 55.5]	57.7	-2.4	-6.2	-1.6	-4.2	8.99	0.55	0.90	138	0.41/0.45/0.320
Strip	BEGV_adv	2e-06	43.3	[33.0, 55.8]	57.4	-3.5	-5.2	-1.5	-3.9	8.93	0.55	0.90	138	0.41/0.44/0.260
Strip	BEGV_adv_cal	2e-06	43.8	[32.1, 57.1]	57.4	-2.8	-5.2	-1.5	-4.1	9.06	0.55	0.90	143	0.40/0.44/0.260
Strip	Oracle	2e-06	43.4	[31.9, 56.6]	57.1	-3.0	-5.0	-1.5	-4.0	9.22	0.52	0.90	143	0.40/0.44/0.260
ImpactAware	Symmetric	2e-06	34.5	[22.6, 50.0]	64.1	-10.2	-6.0	-5.1	-8.3	11.53	0.27	0.71	403	0.43/0.45/0.390
ImpactAware	BEGV	2e-06	45.2	[36.0, 55.6]	57.5	-3.4	-5.9	-0.9	-2.2	8.96	0.57	0.94	78	0.41/0.45/0.330
ImpactAware	BEGV_adv	2e-06	45.6	[36.5, 57.3]	57.7	-4.2	-5.0	-0.8	-2.0	8.95	0.58	0.94	77	0.41/0.44/0.270
ImpactAware	BEGV_adv_cal	2e-06	47.3	[37.6, 59.3]	57.6	-2.1	-5.2	-0.8	-2.2	9.03	0.59	0.94	78	0.41/0.43/0.270
ImpactAware	Oracle	2e-06	46.6	[34.6, 59.6]	57.2	-2.6	-5.1	-0.8	-2.1	9.24	0.56	0.94	78	0.41/0.44/0.260
FrontStack	Symmetric	5e-06	35.4	[20.0, 53.8]	63.8	-12.6	-5.8	-5.2	-4.9	11.59	0.27	0.71	413	0.43/0.45/0.390
FrontStack	BEGV	5e-06	47.1	[37.1, 58.4]	53.5	-0.2	-5.5	-0.3	-0.4	7.84	0.79	0.98	26	0.39/0.44/0.300
FrontStack	BEGV_adv	5e-06	46.4	[35.2, 57.5]	53.3	-1.8	-4.6	-0.3	-0.4	7.86	0.78	0.98	25	0.39/0.43/0.240
FrontStack	BEGV_adv_cal	5e-06	47.8	[38.7, 57.9]	53.2	-0.5	-4.4	-0.2	-0.3	7.80	0.81	0.98	24	0.39/0.43/0.240
FrontStack	Oracle	5e-06	47.7	[39.6, 57.4]	53.2	-0.3	-4.5	-0.3	-0.3	7.96	0.79	0.98	29	0.38/0.43/0.240
Strip	Symmetric	5e-06	16.9	[-3.0, 39.1]	63.4	-17.6	-7.3	-6.5	-15.1	11.60	0.14	0.63	514	0.43/0.45/0.390
Strip	BEGV	5e-06	45.1	[35.1, 56.0]	53.6	-0.8	-5.4	-0.6	-1.7	7.83	0.76	0.96	59	0.39/0.44/0.300
Strip	BEGV_adv	5e-06	45.2	[33.7, 56.9]	53.4	-1.3	-4.9	-0.6	-1.5	7.80	0.78	0.96	56	0.38/0.44/0.260
Strip	BEGV_adv_cal	5e-06	46.3	[36.3, 57.1]	53.4	-0.2	-4.8	-0.6	-1.5	7.74	0.80	0.96	59	0.38/0.45/0.260
Strip	Oracle	5e-06	48.1	[38.9, 58.5]	53.2	1.6	-4.5	-0.7	-1.5	7.99	0.79	0.95	64	0.38/0.44/0.260
ImpactAware	Symmetric	5e-06	28.0	[13.4, 46.0]	64.1	-14.1	-6.6	-5.1	-10.2	11.57	0.22	0.70	405	0.43/0.45/0.390
ImpactAware	BEGV	5e-06	46.4	[36.7, 57.3]	53.7	-0.7	-5.4	-0.3	-0.9	7.77	0.79	0.98	25	0.39/0.43/0.300
ImpactAware	BEGV_adv	5e-06	46.0	[36.5, 57.1]	53.2	-1.5	-4.6	-0.2	-0.9	7.76	0.78	0.98	24	0.39/0.44/0.260
ImpactAware	BEGV_adv_cal	5e-06	46.5	[36.3, 57.7]	53.4	-1.3	-4.5	-0.3	-0.9	7.87	0.78	0.98	25	0.39/0.44/0.260
ImpactAware	Oracle	5e-06	48.5	[37.1, 60.9]	53.1	1.3	-4.7	-0.3	-0.9	7.92	0.82	0.98	27	0.39/0.43/0.260

Simulator: each quoter minus Symmetric on the same seeds (paired bootstrap 95% CI over seeds):

hedger						
FrontStack	2e-07	BEGV	+4.8	[-4.1, +14.3]	+13.7%	+0.07
FrontStack	2e-07	BEGV_adv	+8.6	[-1.0, +18.3]	+24.2%	+0.10
FrontStack	2e-07	BEGV_adv_cal	+9.7	[+1.7, +18.0]	+27.3%	+0.11
FrontStack	2e-07	Oracle	+8.6	[-0.3, +17.4]	+24.4%	+0.10
FrontStack	5e-07	BEGV	+8.2	[-1.3, +17.7]	+23.2%	+0.14
FrontStack	5e-07	BEGV_adv	+8.1	[-1.4, +17.7]	+22.8%	+0.13
FrontStack	5e-07	BEGV_adv_cal	+9.4	[+1.4, +17.6]	+26.5%	+0.15
FrontStack	5e-07	Oracle	+9.8	[+1.3, +18.7]	+27.7%	+0.14
FrontStack	1e-06	BEGV	+10.4	[+1.5, +19.2]	+29.4%	+0.21
FrontStack	1e-06	BEGV_adv	+10.3	[+3.2, +17.1]	+29.0%	+0.21
FrontStack	1e-06	BEGV_adv_cal	+12.9	[+6.5, +19.2]	+36.3%	+0.24
FrontStack	1e-06	Oracle	+12.6	[+4.5, +20.8]	+35.6%	+0.24
FrontStack	2e-06	BEGV	+11.2	[-0.4, +22.9]	+31.6%	+0.32
FrontStack	2e-06	BEGV_adv	+10.8	[+1.7, +19.8]	+30.6%	+0.30
FrontStack	2e-06	BEGV_adv_cal	+12.6	[+3.7, +21.6]	+35.6%	+0.32
FrontStack	2e-06	Oracle	+12.1	[+2.8, +21.8]	+34.3%	+0.30
FrontStack	5e-06	BEGV	+11.7	[-0.4, +23.5]	+33.1%	+0.52
FrontStack	5e-06	BEGV_adv	+10.9	[-1.5, +23.1]	+30.9%	+0.51
FrontStack	5e-06	BEGV_adv_cal	+12.4	[+0.7, +23.9]	+35.0%	+0.54

FrontStack	5e-06	Oracle	+12.3	[+0.3, +24.7]	+34.8%	+0.52
Strip	2e-07	BEGV	+12.8	[+4.9, +20.1]	+75.6%	+0.12
Strip	2e-07	BEGV_adv	+15.8	[+9.2, +22.5]	+93.3%	+0.13
Strip	2e-07	BEGV_adv_cal	+17.2	[+10.5, +24.0]	+101.5%	+0.15
Strip	2e-07	Oracle	+17.9	[+11.3, +24.8]	+105.7%	+0.15
Strip	5e-07	BEGV	+18.0	[+9.0, +26.6]	+106.5%	+0.19
Strip	5e-07	BEGV_adv	+19.7	[+11.9, +26.8]	+116.3%	+0.19
Strip	5e-07	BEGV_adv_cal	+20.6	[+12.6, +28.3]	+121.8%	+0.20
Strip	5e-07	Oracle	+21.0	[+13.1, +28.2]	+123.8%	+0.20
Strip	1e-06	BEGV	+21.6	[+11.6, +30.4]	+127.4%	+0.27
Strip	1e-06	BEGV_adv	+22.5	[+13.6, +31.1]	+132.7%	+0.27
Strip	1e-06	BEGV_adv_cal	+25.3	[+16.6, +33.9]	+149.7%	+0.31
Strip	1e-06	Oracle	+24.0	[+15.5, +31.7]	+142.0%	+0.28
Strip	2e-06	BEGV	+26.5	[+14.2, +37.4]	+156.5%	+0.41
Strip	2e-06	BEGV_adv	+26.4	[+14.8, +36.6]	+155.7%	+0.40
Strip	2e-06	BEGV_adv_cal	+26.9	[+16.3, +36.6]	+158.7%	+0.40
Strip	2e-06	Oracle	+26.5	[+15.1, +37.3]	+156.5%	+0.38
Strip	5e-06	BEGV	+28.1	[+14.6, +39.9]	+166.2%	+0.61
Strip	5e-06	BEGV_adv	+28.2	[+16.3, +38.9]	+166.8%	+0.64
Strip	5e-06	BEGV_adv_cal	+29.4	[+16.4, +40.9]	+173.4%	+0.66
Strip	5e-06	Oracle	+31.2	[+17.3, +43.4]	+184.4%	+0.64
ImpactAware	2e-07	BEGV	+3.8	[-5.0, +12.6]	+10.2%	+0.07
ImpactAware	2e-07	BEGV_adv	+7.2	[-1.2, +14.8]	+19.3%	+0.09
ImpactAware	2e-07	BEGV_adv_cal	+8.3	[+1.2, +15.1]	+22.2%	+0.10
ImpactAware	2e-07	Oracle	+8.9	[+0.7, +17.1]	+23.8%	+0.10
ImpactAware	5e-07	BEGV	+6.6	[-1.6, +14.5]	+17.7%	+0.13
ImpactAware	5e-07	BEGV_adv	+7.7	[+0.8, +14.4]	+20.7%	+0.13
ImpactAware	5e-07	BEGV_adv_cal	+10.2	[+4.0, +16.6]	+27.3%	+0.15
ImpactAware	5e-07	Oracle	+9.6	[+3.2, +16.0]	+25.6%	+0.13
ImpactAware	1e-06	BEGV	+7.2	[+0.5, +13.8]	+19.2%	+0.17
ImpactAware	1e-06	BEGV_adv	+9.2	[+2.3, +15.6]	+24.5%	+0.20
ImpactAware	1e-06	BEGV_adv_cal	+10.2	[+4.9, +15.5]	+27.1%	+0.21
ImpactAware	1e-06	Oracle	+9.5	[+3.0, +15.8]	+25.5%	+0.19
ImpactAware	2e-06	BEGV	+10.7	[+2.7, +18.3]	+31.2%	+0.30
ImpactAware	2e-06	BEGV_adv	+11.2	[+3.7, +18.4]	+32.4%	+0.30
ImpactAware	2e-06	BEGV_adv_cal	+12.8	[+5.9, +19.8]	+37.2%	+0.31
ImpactAware	2e-06	Oracle	+12.1	[+3.1, +20.6]	+35.2%	+0.29
ImpactAware	5e-06	BEGV	+18.4	[+8.3, +28.4]	+65.7%	+0.57
ImpactAware	5e-06	BEGV_adv	+17.9	[+9.2, +26.8]	+64.0%	+0.56
ImpactAware	5e-06	BEGV_adv_cal	+18.4	[+9.7, +27.7]	+65.8%	+0.55
ImpactAware	5e-06	Oracle	+20.4	[+11.6, +30.5]	+72.9%	+0.60

Simulator: each hedger minus FrontStack on the same seeds:

						quoter
BEGV	2e-07	Strip	-10.5	[-13.8, -7.3]	-9.2	10.84 vs 10.88
BEGV	2e-07	ImpactAware	+0.9	[-3.2, +5.3]	-1.3	10.77 vs 10.88
BEGV	5e-07	Strip	-8.7	[-11.6, -5.8]	-7.8	10.37 vs 10.45
BEGV	5e-07	ImpactAware	+0.3	[-3.6, +5.0]	-1.3	10.34 vs 10.45
BEGV	1e-06	Strip	-7.3	[-11.7, -2.9]	-6.1	9.84 vs 9.86
BEGV	1e-06	ImpactAware	-1.1	[-4.5, +2.3]	-1.2	9.81 vs 9.86
BEGV	2e-06	Strip	-3.2	[-7.4, +0.9]	-3.9	8.99 vs 9.00
BEGV	2e-06	ImpactAware	-1.4	[-4.3, +1.1]	-1.2	8.96 vs 9.00
BEGV	5e-06	Strip	-2.1	[-2.9, -0.9]	-1.6	7.83 vs 7.84
BEGV	5e-06	ImpactAware	-0.7	[-2.1, +0.9]	-0.5	7.77 vs 7.84
BEGV_adv	2e-07	Strip	-11.3	[-14.0, -8.9]	-9.1	10.90 vs 10.97
BEGV_adv	2e-07	ImpactAware	+0.6	[-2.5, +4.3]	-1.6	10.89 vs 10.97
BEGV_adv	5e-07	Strip	-6.9	[-10.2, -3.8]	-7.4	10.48 vs 10.40
BEGV_adv	5e-07	ImpactAware	+1.6	[-1.5, +4.9]	-1.2	10.38 vs 10.40
BEGV_adv	1e-06	Strip	-6.3	[-9.9, -2.4]	-5.6	9.83 vs 9.72

BEGV_adv	1e-06	ImpactAware	+1.0	[-3.1, +6.0]	-0.9	9.77 vs 9.72
BEGV_adv	2e-06	Strip	-3.0	[-6.1, +0.3]	-3.6	8.93 vs 9.04
BEGV_adv	2e-06	ImpactAware	-0.6	[-3.8, +2.5]	-1.1	8.95 vs 9.04
BEGV_adv	5e-06	Strip	-1.2	[-2.8, +0.2]	-1.4	7.80 vs 7.86
BEGV_adv	5e-06	ImpactAware	-0.4	[-2.7, +2.5]	-0.5	7.76 vs 7.86
BEGV_adv_cal	2e-07	Strip	-11.0	[-14.3, -7.6]	-8.7	10.83 vs 10.88
BEGV_adv_cal	2e-07	ImpactAware	+0.6	[-2.4, +3.9]	-1.3	10.83 vs 10.88
BEGV_adv_cal	5e-07	Strip	-7.2	[-9.5, -5.0]	-7.3	10.46 vs 10.40
BEGV_adv_cal	5e-07	ImpactAware	+2.8	[-0.4, +6.4]	-1.2	10.39 vs 10.40
BEGV_adv_cal	1e-06	Strip	-6.0	[-9.8, -2.6]	-5.8	9.76 vs 9.73
BEGV_adv_cal	1e-06	ImpactAware	-0.6	[-3.4, +2.6]	-1.1	9.78 vs 9.73
BEGV_adv_cal	2e-06	Strip	-4.2	[-7.9, -0.5]	-4.0	9.06 vs 9.05
BEGV_adv_cal	2e-06	ImpactAware	-0.7	[-2.9, +1.2]	-1.4	9.03 vs 9.05
BEGV_adv_cal	5e-06	Strip	-1.5	[-2.9, -0.1]	-1.6	7.74 vs 7.80
BEGV_adv_cal	5e-06	ImpactAware	-1.4	[-3.0, +0.3]	-0.6	7.87 vs 7.80
Oracle	2e-07	Strip	-9.2	[-11.4, -7.0]	-8.9	10.91 vs 11.01
Oracle	2e-07	ImpactAware	+2.2	[-1.3, +6.2]	-1.3	10.91 vs 11.01
Oracle	5e-07	Strip	-7.3	[-10.1, -4.4]	-7.1	10.56 vs 10.55
Oracle	5e-07	ImpactAware	+1.7	[-0.6, +4.3]	-1.1	10.55 vs 10.55
Oracle	1e-06	Strip	-7.1	[-9.6, -4.1]	-5.5	9.92 vs 9.81
Oracle	1e-06	ImpactAware	-1.0	[-2.3, +0.7]	-0.8	9.89 vs 9.81
Oracle	2e-06	Strip	-4.1	[-8.4, +0.2]	-3.9	9.22 vs 9.21
Oracle	2e-06	ImpactAware	-1.0	[-4.8, +2.9]	-1.2	9.24 vs 9.21
Oracle	5e-06	Strip	+0.4	[-1.9, +2.6]	-1.6	7.99 vs 7.96
Oracle	5e-06	ImpactAware	+0.7	[-3.4, +4.5]	-0.5	7.92 vs 7.96
Symmetric	2e-07	Strip	-18.5	[-29.7, -10.4]	-11.6	11.60 vs 11.59
Symmetric	2e-07	ImpactAware	+1.9	[-3.6, +8.2]	-2.3	11.47 vs 11.59
Symmetric	5e-07	Strip	-18.5	[-29.7, -10.4]	-11.6	11.60 vs 11.59
Symmetric	5e-07	ImpactAware	+1.9	[-3.6, +8.2]	-2.3	11.47 vs 11.59
Symmetric	1e-06	Strip	-18.5	[-29.7, -10.4]	-11.6	11.60 vs 11.59
Symmetric	1e-06	ImpactAware	+2.1	[-3.4, +8.2]	-2.3	11.48 vs 11.59
Symmetric	2e-06	Strip	-18.5	[-29.7, -10.4]	-11.6	11.60 vs 11.59
Symmetric	2e-06	ImpactAware	-0.9	[-5.7, +4.2]	-3.3	11.53 vs 11.59
Symmetric	5e-06	Strip	-18.5	[-29.7, -10.4]	-11.6	11.60 vs 11.59
Symmetric	5e-06	ImpactAware	-7.4	[-10.8, -3.9]	-5.3	11.57 vs 11.59

Mark-outs at 30 minutes against the dealer ($/MMBtu per contract, positive = the client was right), gamma 1e-06, hedger ImpactAware:

						quoter	
Symmetric	consumer	normal	433	0.0191	+0.0000	+0.0001	+0.0009
Symmetric	consumer	pre-release	5	0.0191	-0.0009	+0.0660	+0.0515
Symmetric	producer	normal	114	0.0213	-0.0001	-0.0004	+0.0015
Symmetric	producer	pre-release	2	0.0213	-0.0023	+0.0649	+0.1373
Symmetric	fund	normal	235	0.0070	+0.0012	+0.0011	+0.0022
Symmetric	fund	pre-release	5	0.0070	-0.0030	+0.0625	+0.1131
BEGV	consumer	normal	433	0.0182	-0.0001	-0.0003	-0.0001
BEGV	consumer	pre-release	5	0.0168	-0.0005	+0.0975	+0.1034
BEGV	producer	normal	111	0.0202	-0.0004	-0.0002	+0.0015
BEGV	producer	pre-release	2	0.0171	-0.0023	+0.0652	+0.1375
BEGV	fund	normal	203	0.0062	+0.0006	-0.0006	-0.0004
BEGV	fund	pre-release	7	0.0059	-0.0040	+0.0328	+0.0699
BEGV_adv	consumer	normal	432	0.0183	-0.0002	-0.0003	-0.0002
BEGV_adv	consumer	pre-release	5	0.0166	-0.0005	+0.0975	+0.1034
BEGV_adv	producer	normal	113	0.0202	-0.0002	-0.0001	+0.0018
BEGV_adv	producer	pre-release	2	0.0172	-0.0023	+0.0652	+0.1375
BEGV_adv	fund	normal	166	0.0076	+0.0003	-0.0017	-0.0013
BEGV_adv	fund	pre-release	3	0.0081	-0.0054	+0.0354	+0.0738
BEGV_adv_cal	consumer	normal	432	0.0183	-0.0002	-0.0003	-0.0002
BEGV_adv_cal	consumer	pre-release	5	0.0179	-0.0005	+0.0975	+0.1034
BEGV_adv_cal	producer	normal	113	0.0202	-0.0002	-0.0001	+0.0018

BEGV_adv_cal	producer	pre-release	2	0.0185	-0.0023	+0.0652	+0.1375
BEGV_adv_cal	fund	normal	166	0.0076	+0.0003	-0.0017	-0.0013
BEGV_adv_cal	fund	pre-release	3	0.0131	-0.0054	+0.0354	+0.0738
Oracle	consumer	normal	435	0.0183	-0.0002	-0.0001	-0.0001
Oracle	consumer	pre-release	5	0.0165	-0.0005	+0.0975	+0.1034
Oracle	producer	normal	112	0.0201	-0.0002	-0.0001	+0.0011
Oracle	producer	pre-release	2	0.0171	-0.0023	+0.0652	+0.1375
Oracle	fund	normal	164	0.0076	+0.0005	-0.0014	-0.0014

Variant tox (40 days x 8 seeds):

											hedger
ImpactAware	Symmetric	1e-06	37.3	[26.1, 52.6]	-5.7	-5.2	-7.2	11.47	0.30	0.70	0
ImpactAware	BEGV_adv_cal	1e-06	47.8	[37.1, 62.1]	-4.6	-1.6	-2.6	9.77	0.50	0.89	0

Variant anticipate (40 days x 8 seeds):

											hedger
ImpactAware	Symmetric	1e-06	39.0	[28.1, 53.5]	-5.2	-5.4	-5.3	10.58	0.37	0.70	0
ImpactAware	BEGV_adv_cal	1e-06	50.4	[40.6, 64.1]	-4.2	-1.9	-2.2	9.22	0.60	0.88	0

Variant impact half (40 days x 8 seeds):

											hedger
FrontStack	BEGV_adv_cal	1e-06	49.3	[38.1, 64.4]	-4.8	-1.0	-0.7	9.67	0.53	0.89	0
ImpactAware	BEGV_adv_cal	1e-06	49.3	[37.8, 64.8]	-4.7	-1.0	-1.4	9.77	0.52	0.90	0

Variant impact double (40 days x 8 seeds):

											hedger
FrontStack	BEGV_adv_cal	1e-06	45.1	[33.1, 60.6]	-4.9	-2.7	-3.2	9.79	0.47	0.89	0
ImpactAware	BEGV_adv_cal	1e-06	44.5	[35.0, 57.4]	-4.8	-2.6	-5.2	9.84	0.46	0.90	0

WTI crude (CL): replay of 480 real days 2024-09-18..2026-09-17 (100 EIA release days), 3 seeds

The front contract follows the realised daily close-to-close change (Brownian bridge intraday, the release-day jump at the release time), the rest of the ladder moves with the factor model; the calendar-aware quoter widens fund quotes in the 30 minutes before a release.

													quoter
Symmetric	ImpactAware	82.2	[71.9, 90.4]	113.1	7.8	-3.0	-15.8	-19.9	11.31	0.65	0.63	0.40	0
BEGV	ImpactAware	92.9	[92.0, 94.0]	110.0	5.6	-2.9	-8.5	-11.2	10.18	0.90	0.77	0.31	0
BEGV_adv	ImpactAware	91.9	[91.4, 92.7]	108.8	4.3	-2.5	-8.0	-10.7	10.12	0.90	0.77	0.24	0
BEGV_adv_cal	ImpactAware	91.2	[90.5, 92.2]	108.6	4.2	-2.6	-8.0	-11.1	10.14	0.89	0.76	0.24	0
Oracle	ImpactAware	91.2	[89.9, 92.1]	108.0	4.4	-2.5	-8.1	-10.7	10.15	0.89	0.76	0.23	0

Replay: each quoter minus Symmetric on the same seeds (paired bootstrap 95% CI over seeds):

						hedger
ImpactAware	5e-07	BEGV	+10.7	[+1.6, +20.8]	+13.0%	+0.25
ImpactAware	5e-07	BEGV_adv	+9.7	[+1.2, +19.5]	+11.8%	+0.25
ImpactAware	5e-07	BEGV_adv_cal	+9.0	[+0.4, +18.6]	+10.9%	+0.24
ImpactAware	5e-07	Oracle	+9.0	[-0.4, +20.2]	+10.9%	+0.24

Fund mark-outs ($/bbl per contract):

						quoter
Symmetric	normal	2873	0.1405	+0.0072	+0.0145	+0.0304
Symmetric	pre-release	121	0.1405	-0.0105	+0.0513	+0.0220
BEGV	normal	2256	0.1653	+0.0080	+0.0160	+0.0389
BEGV	pre-release	97	0.1556	-0.0031	+0.0529	+0.0067
BEGV_adv	normal	1754	0.2023	+0.0119	+0.0158	+0.0451
BEGV_adv	pre-release	76	0.1937	-0.0013	+0.0431	+0.0151
BEGV_adv_cal	normal	1753	0.2023	+0.0129	+0.0174	+0.0495
BEGV_adv_cal	pre-release	53	0.2410	+0.0228	+0.0543	+0.0240
Oracle	normal	1758	0.2020	+0.0120	+0.0193	+0.0515

Henry Hub gas (NG): replay of 480 real days 2024-09-18..2026-09-17 (97 EIA release days), 3 seeds

The front contract follows the realised daily close-to-close change (Brownian bridge intraday, the release-day jump at the release time), the rest of the ladder moves with the factor model; the calendar-aware quoter widens fund quotes in the 30 minutes before a release.

													quoter
Symmetric	ImpactAware	55.1	[41.5, 67.4]	56.1	10.6	-4.1	-3.5	-3.9	10.37	0.53	0.77	0.40	21
BEGV	ImpactAware	54.1	[44.7, 61.6]	54.5	8.3	-4.1	-1.8	-2.8	9.34	0.63	0.89	0.38	20
BEGV_adv	ImpactAware	55.2	[43.8, 62.0]	54.4	8.4	-3.5	-1.6	-2.5	9.40	0.64	0.89	0.30	21
BEGV_adv_cal	ImpactAware	54.5	[46.3, 60.7]	54.3	7.8	-3.4	-1.6	-2.5	9.07	0.67	0.89	0.30	21
Oracle	ImpactAware	54.7	[45.6, 61.4]	54.0	8.1	-3.3	-1.6	-2.5	9.11	0.66	0.88	0.29	21

Replay: each quoter minus Symmetric on the same seeds (paired bootstrap 95% CI over seeds):

						hedger
ImpactAware	5e-07	BEGV	-1.0	[-11.3, +5.0]	-1.8%	+0.10
ImpactAware	5e-07	BEGV_adv	+0.0	[-7.7, +5.4]	+0.0%	+0.11
ImpactAware	5e-07	BEGV_adv_cal	-0.6	[-10.8, +4.8]	-1.1%	+0.14
ImpactAware	5e-07	Oracle	-0.5	[-10.3, +4.8]	-0.8%	+0.14

Fund mark-outs ($/MMBtu per contract):

						quoter
Symmetric	normal	2945	0.0070	+0.0002	+0.0005	+0.0009
Symmetric	pre-release	109	0.0070	+0.0079	+0.0289	+0.0373
BEGV	normal	2753	0.0066	+0.0002	+0.0003	+0.0002
BEGV	pre-release	101	0.0070	+0.0091	+0.0274	+0.0358
BEGV_adv	normal	2206	0.0084	+0.0002	+0.0002	+0.0002
BEGV_adv	pre-release	82	0.0088	+0.0103	+0.0298	+0.0396
BEGV_adv_cal	normal	2201	0.0084	+0.0002	+0.0002	+0.0003
BEGV_adv_cal	pre-release	40	0.0139	+0.0005	+0.0063	+0.0067
Oracle	normal	2204	0.0084	+0.0002	+0.0002	+0.0002

EIA release days: jump share, inventory-surprise nowcast, hedging pressure (EIA history, train before 2020 /

root	release days	other days	mean	move	release [95% CI]	mean	move	other	variance ratio	jump share kappa	surprise nowcast coef	out-of-sample [95% CI]	hedging pressure coef	out-of-sample [95% CI]
CL	1999	7638	0.920 [0.871, 0.974]	0.745	0.93	0.00	-0.145 (n=1626)	+0.079 [-0.017, +0.197	+0.057	-0.082 [-0.227	+0.083			
NG	768	6860	0.099 [0.087, 0.114]	0.045	1.19	0.40	-0.094 (n=400)	-0.056 [-0.153	+0.051	-0.011 [-0.101	+0.077			